Summer Research Notes

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Scatter Plots and Describing Association

The most common display for examining the **association** (or relationship) between two *quantitative* variables is the **scatter plot**. A scatter plot simply assigns one axis to one variable, and the second axis to the other, and plots the values against each other. Some examples are shown on the next page.

When we describe distributions of single variables. We looked for center, spread, shape, unusual features, and context. When describing the relationship between variables, we also want to be sure to mention a few key characteristics.

Describing Association: 4 Things to Look For:

1. **Direction**: In what direction do the values of one variable change with increases in the values of the other variable?
   (a) **Positive Association**: Two variables are said to have positive association if the values of one variable tend to *increase* with the values of the second variable. So larger values of one variable generally correspond to larger values of the second variable. Examples?

   (b) **Negative Association**: Two variables are said to have negative association if the values of one variable tend to increase as the values of the second variable tend to *decrease*. So larger values of one variable generally correspond to smaller values of the second variable. Examples?

   (c) Other Types of Association: Two variables may be associated in both a positive and negative fashion over different ranges of values for the two variables. Examples?

   (d) No Association: Two variables are not associated if there is no apparent association (positive, negative, or otherwise) present in the scatterplot. Examples?

2. **Form**: In what form, or how quickly do the values of one variable change with the values of the other variable?
   (a) **Linear**: The relationship between two quantitative variables is said to be *linear* if they exhibit a straight-line pattern in their scatterplot.

   (b) **Nonlinear**: If the relationship is not linear, we say it is nonlinear. Common forms of nonlinear relationship are *exponential* or *quadratic*.

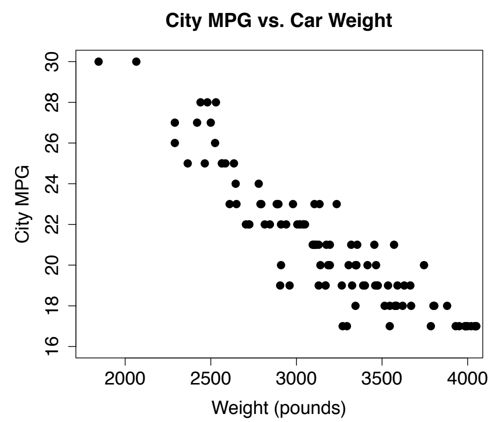
3. **Strength**: How much variability or spread is present in the scatterplot?

   - Lower scatter (stronger association) indicates clearer trends or patterns from which estimation is more precise. If there is too much scatter, we may not be able to discern any relationship between the two variables.

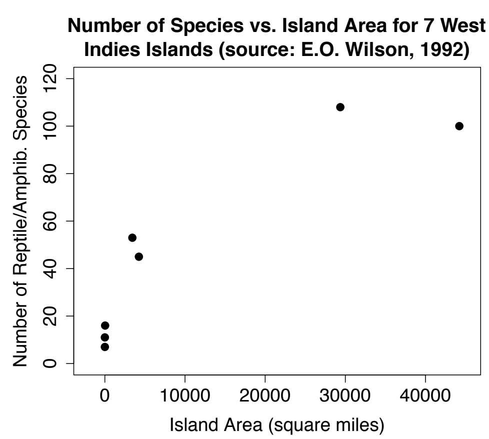
4. **Outliers**: You should note any points which do not fit the overall relationship.

Examples:

1. Is there a relationship between the mileage (MPG) and weight of cars, and if so, what is the nature of the relationship?



2. Is there an association between island size and the number of reptile or amphibian species an island can support? How would you describe this relationship?



Correlation

Correlation describes the *linear* relationship between two quantitative variables. There could be an association between variables that is not linear, and we would say there is not a strong correlation.

Simply looking at a scatterplot to obtain correlation can be subjective, however. There is an objective way to measure correlation by using the **correlation coefficient**. This is also known as the Pearson correlation coefficient.

$$\boxed{r = \frac{1}{n-1} \sum_{i=1}^{n} \left[\left(\frac{x_i - \bar{x}}{s_x} \right) \cdot \left(\frac{y_i - \bar{y}}{s_y} \right) \right]}$$

Correlation interpretation

A very general rule of thumb for interpreting the correlation coefficient is:

If $|r| = 1$, then the correlation is "perfect".

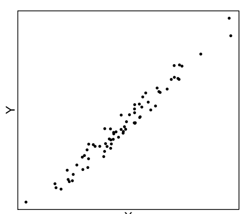
If $0.9 \leq |r| < 1$, then the correlation is "strong".

If $0.6 \leq |r| < 0.9$, then the correlation is "moderate".

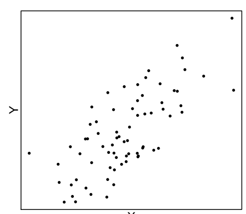
If $0 < |r| < 0.6$, then the correlation is "weak".

If $|r| = 0$, then there is no correlation.

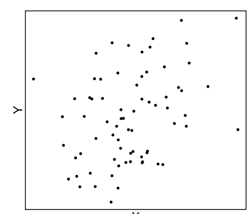
There are other correlation coefficients, like the Kendall and Spearman correlation. In **R**, you can specify the method in the `cor()` function, but we will stick with the Pearson correlation coefficient in this class since it is widely used and the results from all are usually similar.



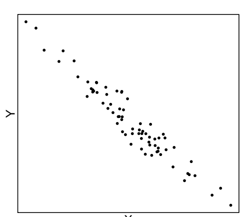
Strong Positive ($r = 0.982$)



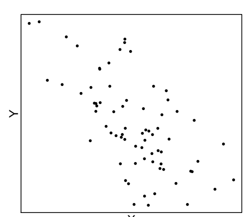
Moderate Positive ($r = 0.724$)



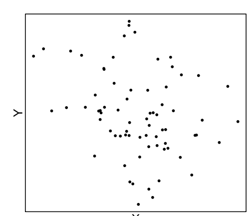
Weak Positive ($r = 0.343$)



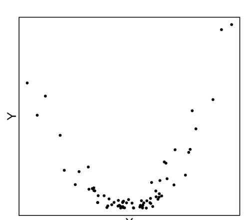
Strong Negative ($r = -0.963$)



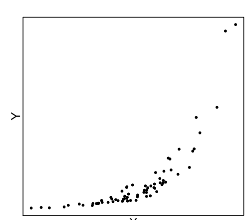
Moderate Negative ($r = -0.639$)



Weak Negative ($r = -0.267$)



Strong Quadratic ($r = 0.209$)



Strong Exponential ($r = 0.770$)

Linear Regression Overview

A linear regression model is one in which we fit a line (in simple linear regression) or a hyperplane (in multiple linear regression) with two primary goals in mind:

- To get predictions of the response variable, Y, using the explanatory variable(s), X (sometimes X_1, X_2, ..., X_p).
- To quantify the nature and the magnitude of the association between Y and the predictor(s). That is, understand what the slope values are telling us.

In general, the regression model can be written

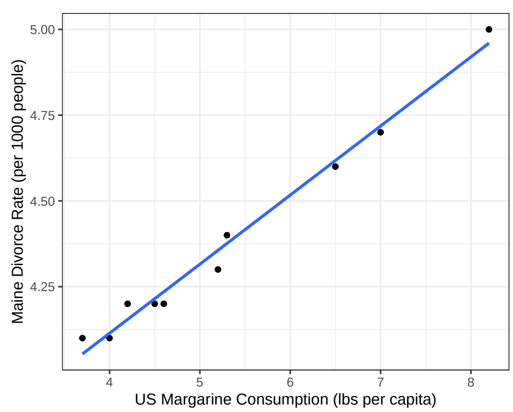
$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

We can have p predictor variables, each of which has it's own slope parameter: the β_j in front of it. In addition, there is an intercept β_0. So, all together, there are $p+1$ parameters in this model. The ε is an error term. If we had data for the entire population, then we could compute these β parameters exactly, but we usually do not. That means we have to estimate them. The estimated parameters are $\hat{\beta}_j$, which are usually written as b_j. So, our predicted model becomes

$$\hat{y}_i = b_0 + b_1 x_{i1} + b_2 x_{i2} + \cdots + b_p x_{ip}$$

The y has a hat on it to indicate this a predicted value of y rather than an actual y value. We compute these estimated parameters by finding the values that minimize the sum of the squared residuals. A residual is the difference between the actual y value and the predicted one. The residuals are written $e_i = y_i - \hat{y}_i$.

Consider an example of attempting to predict the divorce rate in Maine (in divorces per 1000 Mainers) using the margarine consumption of the United States (in pounds per person per year). This data set is called `divorce_margarine` and can be found in the `dslabs` **R** library.



This plot shows (perhaps unexpectedly!) a strong, positive, linear association between the two variables. So these variables would be good candidates for a regression model.

We can fit a regression model in **R** using the `lm()` function. Then we can display lots of useful output using the `summary()` function. Like so:

```
library(dslabs)
maine_data <- divorce_margarine |>
  rename(margarine = margarine_consumption_per_capita,
         divorce = divorce_rate_maine) # Shorten the variable names
maine_mod <- lm(divorce ~ margarine, data = maine_data)
summary(maine_mod)
```

```
Call:
lm(formula = divorce ~ margarine, data = maine_data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.05583 -0.01816 -0.01452  0.03601  0.04625

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.308626   0.048032   68.88 2.20e-12 ***
margarine   0.201386   0.008735   23.05 1.33e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03841 on 8 degrees of freedom
Multiple R-squared:  0.9852,    Adjusted R-squared:  0.9833
F-statistic: 531.5 on 1 and 8 DF,  p-value: 1.33e-08
```

This output tells us that the intercept of the model is 3.308626 and the slope is 0.201386. So, we can write the regression model

$$\hat{y} = 0.2014x + 3.309 \quad \text{or} \quad \hat{y} = 3.309 + 0.2014x$$

or, in the context of the problem:

$$\widehat{\text{divorce rate}} = 0.2014(\text{margarine}) + 3.309$$

Interpretations

Intercept interpretation: The intercept tells us the predicted value of Y when X is equal to zero. In this case, we predict that the divorce rate in Maine will be 0.2014 per 1000 people when the per capita consumption of margarine in the US is zero lbs.

Slope interpretation: The slope is the predicted change in the Y variable when the X variable increases by 1 unit. In this case, as the per capita consumption of margarine in the US increases by 1 lb, the divorce rate in Maine is predicted to increase by 0.2014 per 1000 people. In multiple linear regression, there are more variables in the model, so we need to add the phrase "holding all other variables constant" at the end.

Some other important notes: This output is showing the results of two hypothesis tests, each with their own p-value. The p-values are shown in the `Pr(>|t|)` column. The one in the row for the intercept is testing the hypotheses

$$H_0 : \beta_0 = 0$$
$$H_A : \beta_0 \neq 0$$

In other words, it is testing if the true intercept is different than zero. The one in the row for the variable, `margarine_consumption_per_capita` in this case, is testing the hypotheses

$$H_0 : \beta_1 = 0$$
$$H_A : \beta_1 \neq 0$$

In other words, it is testing if the true slope is different than zero. Since the p-values are very small for both (notice the scientific notation: `2.20e-12` $= 2.20 \times 10^{-12} = 0.0000000000022$), we have extremely strong evidence each parameter is not equal to zero.

Another statistic to pay attention to is the `Multiple R-squared`. This is the **coefficient of determination** and it tells us the proportion of the variation in the Y variable that is being explained by this model. Here, we can say that 98.52% of the variation in the divorce rate in Maine is explained by the per capita consumption of margarine in the US. This is a very high R^2 value and indicates this model fits the data extremely well.

Predictions

We can then get a prediction for the divorce rate by plugging in a value for the margarine variable. For example, the predicted divorce rate in Maine when the margarine consumption in the US is 5 lbs per capita is $\widehat{\text{divorce rate}} = 0.2014(5) + 3.309 = 4.316$ per 1000 people. In **R**, we can do this using the `predict()` function.

```
predict(maine_mod, data.frame(margarine = 5))
```

```
       1
4.315556
```

We can even make a **confidence interval** for the average Y value using the `interval = "confidence"` argument. Or we can make a **prediction interval** for any Y value using the `interval = "prediction"` argument:

```
predict(maine_mod, data.frame(margarine = 5), interval = "confidence")
```

```
       fit      lwr      upr
1 4.315556 4.286815 4.344298
```

```
predict(maine_mod, data.frame(margarine = 5), interval = "prediction")
```

```
       fit      lwr      upr
1 4.315556 4.222435 4.408678
```

These intervals tell us each of the following:

- We are 95% confident that the average divorce rate in Maine for all years when the margarine consumption in the US is 5 lbs is between 4.29 and 4.34 per 1000 people.
- We are 95% confident that the divorce rate in Maine for any year when the margarine consumption in the US is 5 lbs is between 4.22 and 4.41 per 1000 people.

Categorical Predictor Variables

We can also have predictor variables that are categorical. For, instance, suppose we have another variable that indicates whether the year was classified as a recession in the United States. We can add that to our model:

```r
maine_data2 <- maine_data |>
  mutate(recession = c("No", "Yes", "No", "No", "No",
                       "No", "No", "No", "Yes", "Yes"))
maine_mod2 <- lm(divorce ~ margarine + recession, data = maine_data2)
summary(maine_mod2)
```

```

Call:
lm(formula = divorce ~ margarine + recession, data = maine_data2)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.046829 -0.019383 -0.001612  0.023834  0.045122 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.287339   0.047649  68.991 3.53e-11 ***
margarine   0.203358   0.008342  24.376 4.98e-08 ***
recessionYes 0.035981   0.025314   1.421    0.198    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03617 on 7 degrees of freedom
Multiple R-squared:  0.9885,	Adjusted R-squared:  0.9852 
F-statistic: 300.7 on 2 and 7 DF,  p-value: 1.634e-07
```

R creates what is called a **indicator variable** for each category of the variable except one, which it keeps as the baseline category. Here, the `recession` variable has two categories, `No` and `Yes`, but only has one indicator variable: `recessionYes`. This means that `No` is being treated as the baseline. We can interpret the slope of an indicator variable like this:

Indicator variable slope interpretation: The slope is the predicted change in the Y variable when the X variable is changes from the baseline to the category in question. In this model, we can say that the divorce rate in Maine is predicted to be 0.035981 per 1000 people higher in recession years compared to non-recession years, holding margarine consumption per capita constant.

Linear Regression Assumptions and Diagnostics

There are some assumptions that must be satisfied for the model to be fit and then for inference to be performed:

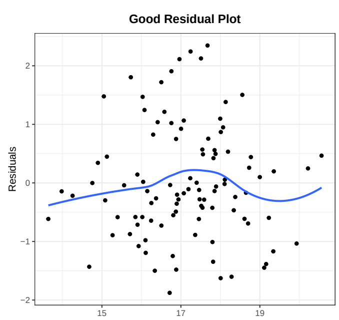
1. The means of the errors at a given value of X must be 0: $E(\varepsilon|x_i) = 0$ for all i. This indicates that the variables have a linear association. This assumption is the only one that is needed for prediction.
2. The variances of the errors at a given value of X must be equal: $\text{Var}(\varepsilon|x_i) = \sigma^2_\varepsilon$ for all i. This assumption is needed for inference.
3. The errors must follow a normal distribution: $\varepsilon \sim N$. This assumption is needed for inference.
4. The errors must be independent of each other. Having a random sample from the population ensures this. This assumption is needed for inference.
5. There is no multicollinearity. That means the predictor variables are not too correlated with each other. This only applied in multiple linear regression. This assumption is needed for inference and interpretation.

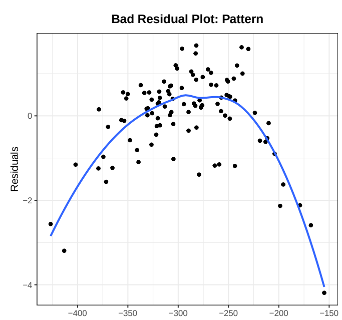
These assumptions are all about the true, population errors, which are unknown. So we never know for sure if these are met, but the reality is that these regression models work well if these are *reasonably* met. The way we assess them is by looking at some **diagnostic plots** like a **residual plot** and a **QQ plot**.

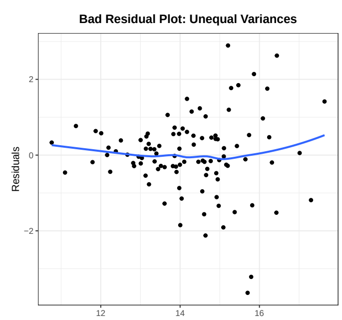
Residual Plots: Residual plots plot the residuals against the fitted values (the $\hat{y}$ values). These can be used to check the first and second assumptions listed above: the linearity and equal variance assumptions. The residuals can be found in **R** using the `residuals()` function. Ideally, the residual plots would show random scatter without any sort of discernible pattern and no fanning. It can be helpful to use `geom_smooth()` to plot a smoothed curve to the plot. This curve should be (mostly) horizontal at $y = 0$. A residual plot code template might look like this:

```
data.frame(resids = residuals(regression_model),
           fitted = fitted.values(regression_model)) |>
  ggplot(aes(x = fitted, y = resids)) +
  geom_point() +
  geom_smooth(method = "loess", formula = "y ~ x", se = F) +
  labs(x = "Fitted Values", y = "Residuals")
```

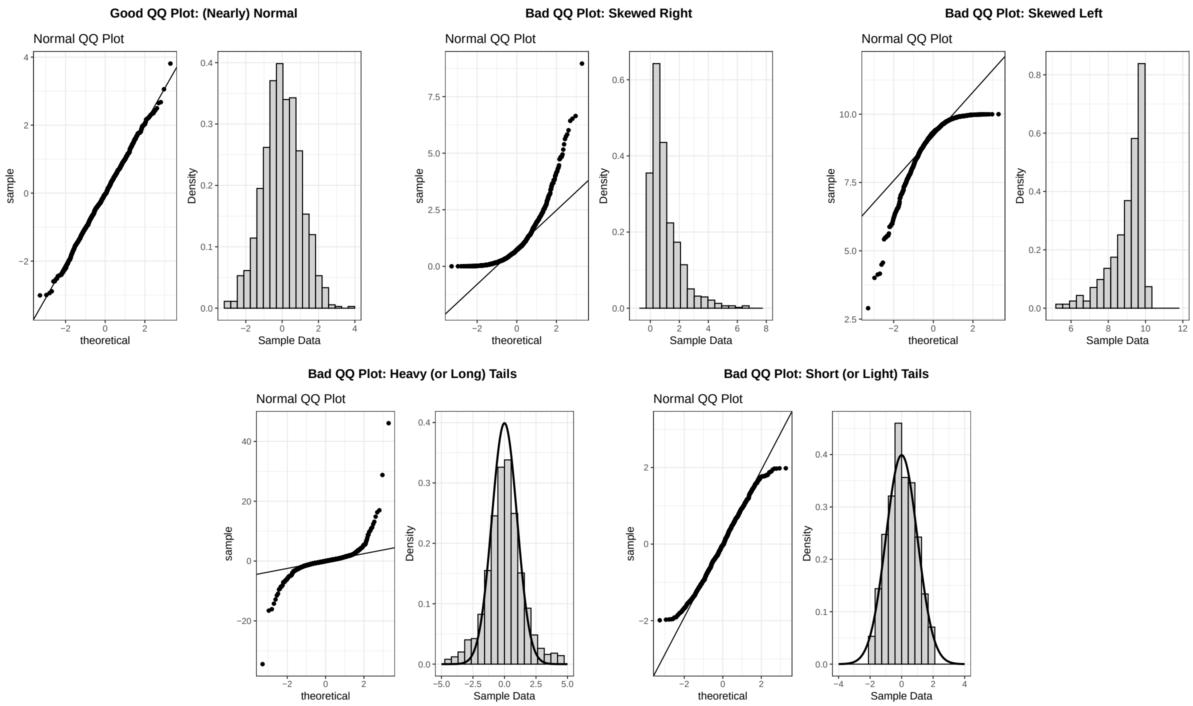
Ideally, the residual plots would show random scatter without any sort of discernible pattern and no fanning. Some examples are shown below.







QQ Plots: QQ plots plot the sample quantiles against the theoretical quantiles. This can be used to check the third assumption above: the normality assumption. If the distribution of the sample (the residuals in our case) are approximately normal, then the points should fall along a straight line. Otherwise, they will deviate from the line. Regression inference is fairly robust against non-normality, so as long as the points are reasonably close to the line, then we will consider this assumption reasonably met.



We can make a QQ plot in **R** using the following code as a template:

```
data.frame(resids = residuals(regression_model)) |>
  ggplot(aes(sample = resids)) +
  geom_qq_line() +
  geom_qq() +
  labs(title = "Normal QQ Plot")
```

What do we do if one or more of the assumptions does not seem to be reasonably satisfied? In that case, it is time for some **remedial measures**, which won't be discussed here.

Influence Statistics

There are also ways of flagging unusual points in a regression model. The most common ones are:

- **Leverage**: Higher leverage points are more unusual in their x-value(s).
- **Cook's Distance**: Points with larger Cook's distance values alter the model's predictions quite a bit when they are added or removed.
- **DfFits**: Points with larger DfFits (difference of fits) values alter the model's prediction for that particular point quite a bit when they are added or removed.
- **DfBetas**: Points with larger DfBetas (difference of betas) values alter the model's estimate for that particular beta (intercept or slope) quite a bit when they are added or removed.

Using `math3150package` to Create Important Plots

I made an **R** package called `math3150package` that has some useful functions in it for getting residual, QQ, and influence plots. You'll encounter this package agin if you take the Math 3150 – Applied Linear Models class. You can install this package by first installing the `pak` package and then using that to install the `math3150package` from GitHub:

```
install.packages("pak")
library(pak)
pak("rbrown53/math3150package")
```

In the `math3150package` library are the `residual_plots()` and `influence_plots()` functions. The `residual_plots()` function makes a residual and QQ plot for the model given and the `influence_plots()` function makes some residual plots and plots for leverage values, Cook's distance, DfFits, and DfBetas values.

```
library(math3150package)
residual_plots(maine_mod, resid_type = "raw")
influence_plots(maine_mod)
```

Any point flagged as potentially influential will be highlighted in red and the observation number will be shown.

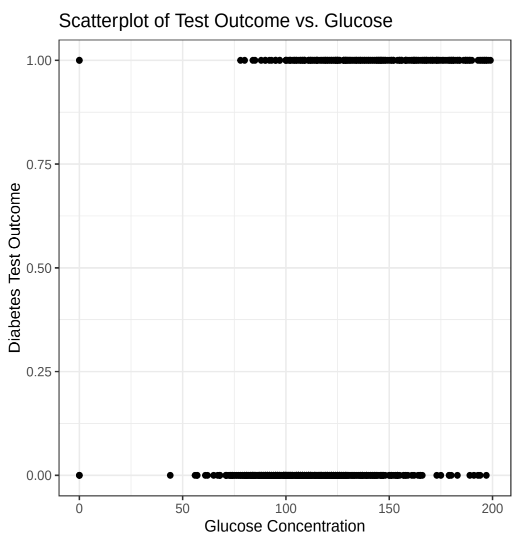
Logistic Regression

As was stated before, the response variable for logistic regression is always either 0 or 1. To see when logistic regression models might be applicable, consider the following example.

Example: The National Institute of Diabetes and Digestive and Kidney Diseases conducted a study on 768 adult female Pima Native Americans living near Phoenix. The purpose of the study was to investigate factors related to diabetes. The response variable (y) in this study (*test*) was whether or not the patient showed signs of diabetes (1 = yes, 0 = no). One explanatory variable (x) considered was the plasma glucose concentration (in mg/dL) at 2 hours in an oral glucose tolerance test (*glucose*). Many other potential explanatory variables were considered as well, but for now, we will focus just on the glucose variable.

- Additionally, we will assume that the subjects used in the study were independently selected. This assumption is critical to the use of logistic regression models. If this assumption is not satisfied (as with time series or spatial data), a class of models known as *auto*logistic models may be used for predicting a binary response.
- The goal here is to predict the diabetes status or probability of showing signs of diabetes given the glucose concentration of the subject.

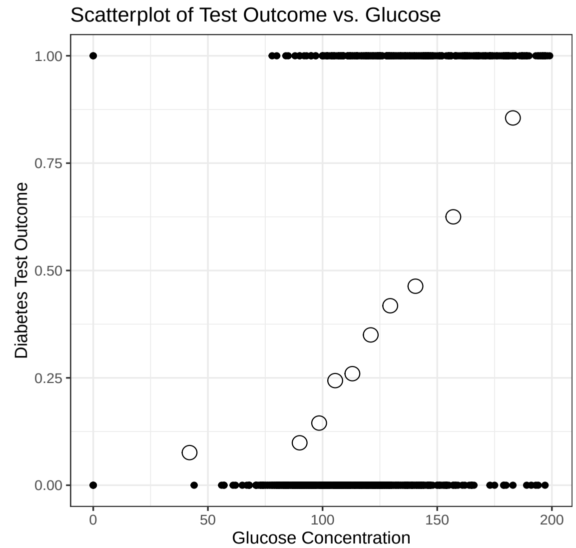
To examine the relationship between the incidence of diabetes and glucose concentration, a scatterplot was constructed below. What does this plot tell you? Anything? Would a linear regression model be appropriate here?



- Better Idea: We could categorize the glucose concentrations into groups and plot the mean diabetes incidence (proportion of subjects in that group testing positive for diabetes) vs. the glucose group. To see how this works, consider the following table and corresponding plot shown below. What do you notice about the shape of the relationship between the proportion of subjects with diabetes and glucose?

Frequency Table of Glucose Group by Diabetes

Glucose Group	n	Diabetes Absent	Diabetes Present	Proportion with Diabetes
0-85	79	73	6	0.076
86-95	81	73	8	0.099
96-102	76	65	11	0.145
103-109	78	59	19	0.244
110-117	77	57	20	0.260
118-125	80	52	28	0.350
126-134	67	39	28	0.418
135-147	82	44	38	0.463
148-167	72	27	45	0.625
168-200	76	11	65	0.855
Total	768	500	268	0.349



- The association between the proportion of subjects with diabetes and glucose level is somewhat S-shaped. This type of relationship is well-represented by the **logistic model**:

$$\pi(x) = P(Y=1|x) = E(Y|x) = \frac{\exp\{\beta_0 + \beta_1 x\}}{1+\exp\{\beta_0 + \beta_1 x\}}, \quad \text{where}$$

  $\pi(x)$ represents the probability of having diabetes given a glucose concentration of x. The portion of this model in the exponent is just a linear function of the parameters β_0, β_1. The **R**-code used to create the table and graph is given in the `Notes 9 Script.R` file.

- The logistic regression model is not the only type of function (known as a link function) used for binary data which has this S-shape. Another commonly considered model for this situation is the **probit model**, a model based on the distribution function of the normal distribution. This will not be discussed here.

Notes:

- $P(Y=0|x) = \dfrac{1}{1+\exp\{\beta_0 + \beta_1 x\}}$. Why?

- Consider the transformation: $\log\left[\dfrac{P(Y=1|x)}{1-P(Y=1|x)}\right] = \log\left[\dfrac{P(Y=1|x)}{P(Y=0|x)}\right]$.
  This is known as the logit transformation, and is interpreted as the **log odds** that $y=1$ (diabetes) versus $y=0$ (no diabetes) given x (glucose). In terms of the example, as glucose concentration (x) increases, we expect the log odds of having diabetes ($y=1$) to increase as well. This logistic model specifies that this increase occur in the form of an S-shape as specified earlier.

- In addition to the simple interpretation of the logit function as a log odds ratio, this transformation has another useful property. Evaluating the logit function:

$$\begin{aligned}\boxed{\text{logit }[P(Y=1|x)]} &= \log\left[\frac{P(Y=1|x)}{P(Y=0|x)}\right] = \log\left[\frac{\exp\{\beta_0+\beta_1 x\}/(1+\exp\{\beta_0+\beta_1 x\})}{1/(1+\exp\{\beta_0+\beta_1 x\})}\right] \\ &= \log\left[\exp\{\beta_0+\beta_1 x\}\right] = \boxed{\beta_0+\beta_1 x}.\end{aligned}$$

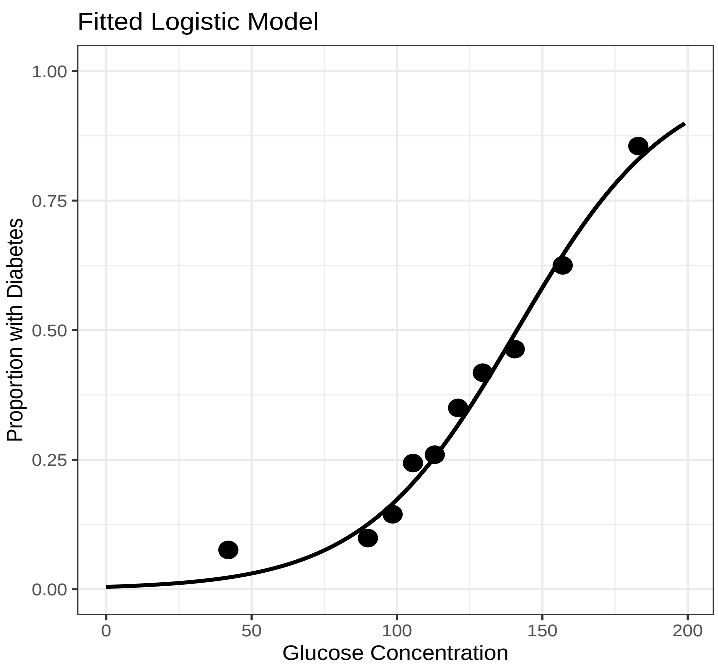
  So the logit function is a linear function of the parameters. Why is this important? This means that on the logit scale, we can appeal to what we know about linear regression to interpret the model parameters.

- The equality $\boxed{\text{logit}[P(Y=1|x)] = \beta_0+\beta_1 x}$ is known as the logistic regression model.

To see how well this model fits the data, reconsider the categorized table and plot shown earlier on the proportion of diabetes subjects versus glucose concentration:

Frequency Table of Glucose Group by Diabetes

Glucose Group	n	Diabetes Absent	Diabetes Present	Mean (Proportion)	Glucose	$\hat{\pi}(x)$
0-85	79	73	6	0.076	42.5	0.023
86-95	81	73	8	0.099	90.0	0.125
96-102	76	65	11	0.145	98.5	0.165
103-109	78	59	19	0.244	105.5	0.205
110-117	77	57	20	0.260	113.0	0.255
118-125	80	52	28	0.350	121.0	0.317
126-134	67	39	28	0.418	129.5	0.390
135-147	82	44	38	0.463	140.5	0.493
148-167	72	27	45	0.625	157.0	0.645
168-200	76	11	65	0.855	183.0	0.829
Total	768	500	268	0.349		



$$\hat{\pi}(x) = \frac{\exp\{-5.35+.0379x\}}{1+\exp\{-5.35+.0379x\}}$$

So, the proposed model for these data is the logistic regression model given by:

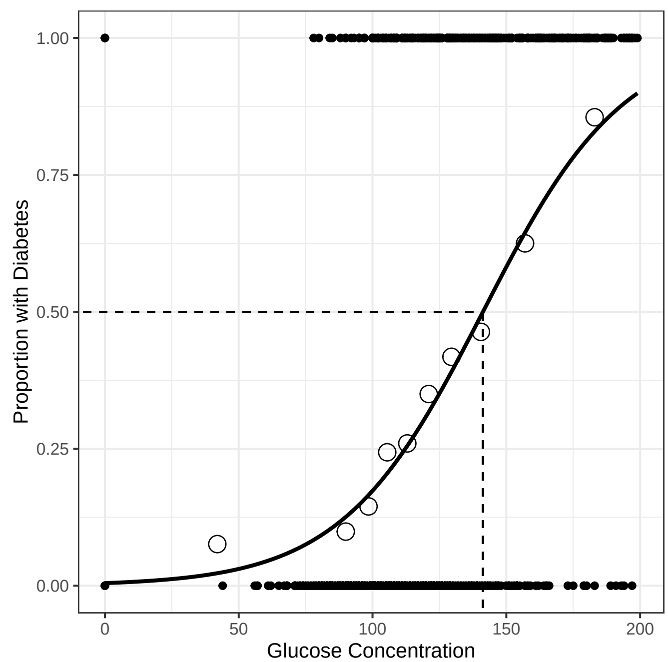
$$\pi(x) = \frac{\exp\{\beta_0 + \beta_1 x\}}{1 + \exp\{\beta_0 + \beta_1 x\}} \qquad \text{or} \qquad \hat{\pi}(x) = \frac{\exp\{b_0 + b_1 x\}}{1 + \exp\{b_0 + b_1 x\}},$$

where x = glucose concentration, and $\pi(x) = E(Y|x)$ = the probability of diabetes at glucose concentration x. To fit the logistic regression model in **R**, we use the `glm()` function with the option `family = binomial()` to select binomial (*logistic*) regression. The code to produce the logistic regression fit is given below along with the corresponding output.

```
# Fits a logistic regression model of diabetes presence vs. glucose
logitmod <- glm(test ~ glucose, data = pima, family = binomial())
summary(logitmod) # Summary of logistic fit
```

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.350080   0.420827  -12.71   <2e-16 ***
glucose      0.037873   0.003252   11.65   <2e-16 ***

    Null deviance: 993.48  on 767  degrees of freedom
Residual deviance: 808.72  on 766  degrees of freedom
AIC: 812.72
```

The resulting logistic regression equation is thus: $\hat{\pi}(x) = \dfrac{\exp\{-5.35 + 0.0379x\}}{1 + \exp\{-5.35 + 0.0379x\}}$. This model is plotted here:



Commonly Desired Quantity:
One quantity commonly computed with logistic regression models is that value of x such that the probability that $y = 1$ is exactly equal to 0.50. This is called the LD_{50} value. In terms of our problem, it represents the glucose concentration at which your probability of having diabetes becomes greater than 50%. At the LD_{50} point, say x^*, we know that $\pi(x^*) = 0.5$. This will occur when $\exp(\beta_0 + \beta_1 x^*) = 1$ which is when $\beta_0 + \beta_1 x^* = 0$.

We can estimate this by solving $b_0 + b_1 x^* = 0$

$$-5.35 + 0.0379x^* = 0 \;\Rightarrow\; x^* = \underline{141.26}.$$

Prediction: Suppose we want to predict the probability that a member of this population tests positive for diabetes if her glucose concentration level is 100. This can be done with the following **R**-code:

```
# Computes the prediction at x=100 on the logit scale (inside the exp)
logitpred <- predict(logitmod, data.frame(glucose = 100), se.fit = TRUE)
ilogit(logitpred$fit) # Predicted prob. at x=100
```

```
        1
0.1732486
```

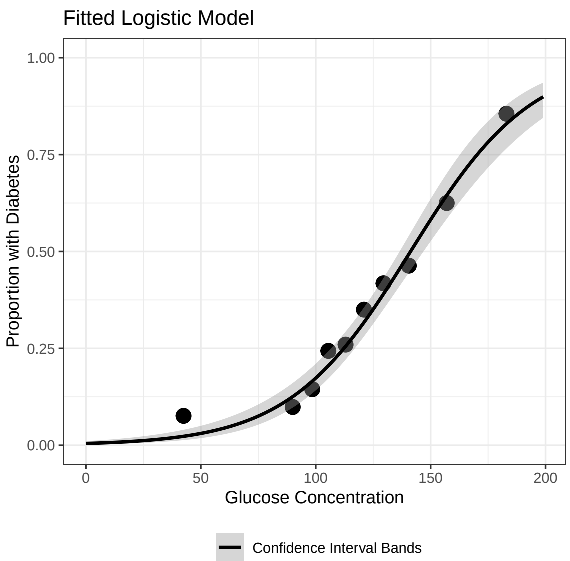
```
se = logitpred$se.fit # SE of prediction
ilogit(c(logitpred$fit-1.96*se, logitpred$fit+1.96*se)) # CI for prediction
```

```
        1         1
0.1415424 0.2103170
```

giving 0.173 as the estimated probability of a positive diabetes test with a 95% confidence interval for the probability given by $(0.142, 0.210)$.

We can extend this idea to get confidence intervals for the true probability for any glucose value in the range of our data.

```
ggplot(pima, aes(x = glucose, y = test)) +
  labs(x = "Glucose Concentration", y = "Proportion with Diabetes",
       title = "Fitted Logistic Model") +
  geom_point(data = pima_props, aes(x = midpoints, y = props), size = 4) +
  geom_smooth(method = "glm", method.args = list(family = "binomial"),
              formula = "y ~ x", se = T, color = "black",
              aes(fill = "Confidence Interval Bands")) +
  lims(y = c(0, 1)) +
  theme_bw() +
  scale_fill_manual("", values = "gray60") +
  theme(legend.position = "bottom")
```



How *well* does the model predict? To address this, we could compute the predicted probabilities of a positive test for all 768 women in the study and then using 0.5 as a cutoff, see how well these predicted probabilities agree with the actual test results. This was done in **R** producing the following contingency table. How well does the model do?

```
# Binary vector set to 1 if the prob. exceeds 0.50 and 0 otherwise
ptest <- as.numeric(logitmod$fitted > 0.5)
table(truth = pima$test, prediction = ptest) # Contingency table of outcomes
```

In this case, the model correctly predicts $(443 + 130)/768 = 74.6\%$ of the cases using 0.5 as a cutoff. Other cutoffs can be used as well. A common visual technique for identifying such a cutoff is the receiver operating characteristic (ROC) curve (not considered here).

```
      prediction
truth   0   1
    0 443  57
    1 138 130
```

- Since we are using our data to predict the outcomes of our data, this accuracy level is somewhat misleading and will tend to be optimistic. To avoid this, with a sufficient amount of data, it is recommended to first remove a subset of your data, and treat the remaining data as a training set. We then use the training set to classify the removed observations and estimate the classification accuracy for this test data set. In this way, the data used to construct the model are independent of the data used to estimate the model accuracy.
- Cutoff values others than 0.5 can also be used. If it is worse to incorrectly predict a 1 than a 0, it makes sense to increase the cutoff and vise-versa.

Interpretation of the Logistic Regression Coefficients

The estimate of β_1 was $b_1 = 0.0379$. How do we interpret this value?

For the model $\hat{\pi}(x) = \dfrac{\exp\{b_0 + b_1 x\}}{1 + \exp\{b_0 + b_1 x\}}$, the "slope" is interpreted in terms of the odds ratio (this is explained a couple pages from now). Namely, $\exp(b_1)$ is the estimated odds of a success at the new value divided by the estimated odds of success at the previous value. So,

$$\widehat{\text{Odds Ratio}} = \widehat{OR} = \frac{\text{odds}_{\text{new}}}{\text{odds}_{\text{prev}}} = \frac{\frac{\hat{\pi}(x+1)}{1-\hat{\pi}(x+1)}}{\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)}} = \exp(b_1).$$

<u>Example</u>: In the diabetes example we saw that $\hat{\pi}(100) \approx \frac{\exp\{-5.35+0.0379(100)\}}{1+\exp\{-5.35+0.0379(100)\}} \approx 0.173249$ and $\hat{\pi}(99) = \frac{\exp\{-5.35+0.0379(99)\}}{1+\exp\{-5.35+0.0379(99)\}} \approx 0.167891$.

So $\text{odds}_{\text{new}} = \frac{0.173249}{1-0.173249} = 0.20955$ and $\text{odds}_{\text{prev}} = \frac{0.167891}{1-0.167891} = 0.20177$ and thus $\widehat{OR} = 0.20955/0.20177 = 1.0386 = \exp(0.0379)$.

When x increases by c units, the predicted change in the odds of a success is $\exp(c \cdot b_1)$.

Interpreting the intercept:
The inverse logit of the intercept of a logistic regression model gives the predicted probability of a success when all predictor variables equal zero. In the diabetes example, b_0 is -5.35. Taking the inverse logit of the intercept gives $\hat{\pi}(0) = \dfrac{\exp(-5.35)}{1+\exp(-5.35)} = 0.0047$. This means

Interpreting the slope of an indicator variable:
Consider the smoking example from the beginning of this set of notes again. Let's fit a logistic regression model using cancer as Y and smoking as X:

```
cancer <- c(rep(1,1357),rep(0,1357))             # Define the cancer vector
smoking <- as.factor(c(rep(1, 1350), rep(0, 7),  # Define the smoking vector
                       rep(1, 1296), rep(0, 61)))
cancer_mod <- glm(cancer ~ smoking, family = binomial()) # Fit the model
summary(cancer_mod)
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.1650     0.3990  -5.425 5.78e-08 ***
smoking1      2.2058     0.4009   5.502 3.76e-08 ***
```

As before, when x increases by 1, the predicted change in the odds of a success is $\exp(b_1)$. That means

Alternative Slope Interpretation (Using Derivative)

Recall that $\hat{\pi}(x) = \dfrac{\exp(b_0+b_1x)}{1+\exp(b_0+b_1x)} = \dfrac{e^{b_0+b_1x}}{1+e^{b_0+b_1x}}$. Let's use the derivative to find the rate of change at a given value of x. Using both the quotient rule and chain rule:

$$\begin{aligned}\frac{d\hat{\pi}}{dx} &= \frac{(b_1e^{b_0+b_1x})(1+e^{b_0+b_1x}) - (e^{b_0+b_1x})(b_1e^{b_0+b_1x})}{(1+e^{b_0+b_1x})^2} \\ &= \frac{b_1e^{b_0+b_1x} + (b_1e^{b_0+b_1x})(e^{b_0+b_1x}) - (e^{b_0+b_1x})(b_1e^{b_0+b_1x})}{(1+e^{b_0+b_1x})^2} \\ &= \frac{b_1e^{b_0+b_1x}}{(1+e^{b_0+b_1x})^2} = b_1 \cdot \frac{e^{b_0+b_1x}}{1+e^{b_0+b_1x}} \cdot \frac{1}{1+e^{b_0+b_1x}}\end{aligned}$$

Now, $\dfrac{e^{b_0+b_1x}}{1+e^{b_0+b_1x}} = \hat{\pi}(x)$ and $\dfrac{1}{1+e^{b_0+b_1x}} = 1-\hat{\pi}(x)$. That means:

$$\frac{d\hat{\pi}}{dx} = b_1 \cdot \hat{\pi}(x) \cdot (1-\hat{\pi}(x)).$$

Therefore, the rate of change in the predicted probability of a "success" at x is $b_1 \cdot \hat{\pi}(x) \cdot (1-\hat{\pi}(x))$.

Odds and Odds Ratio

Logistic regression models are regression models for binary response variables. Before we get into how these work, we should quickly discuss the **odds** of an event and the **odds ratio** of two events.

For a probabilistic event, the **odds** are used to compare the chances of that event happening with the chances of that event not happening.

The odds of an outcome can be found by dividing the number of outcomes in one category to the number of outcomes not in that category. It can also be found by taking the probability of an event and dividing it by 1 minus the probability of that event:

$$\text{odds}(A) = \frac{P(A)}{1 - P(A)} \text{ “to 1”}$$

- If the odds > 1 to 1 then the outcome of interest is more likely than not.
- If the odds < 1 to 1 the outcome of interest is not as likely as everything else.
- odds $= 1$ to 1 implies an even split between the outcome of interest and every other possible outcome.

We say “to 1”, because the odds compares the likelihood of an event happening to that event not happening. So, if the odds of the Broncos beating the Browns are listed as “4 to 1”, we could say that we would expect the Broncos to beat the Browns 4 times for every time the Browns beat the Broncos.

Example: What are the odds that when flipping two coins, both come up heads?

Translating from odds to probability:
If the odds of an event are $\text{odds}(A) = a$ to b, then $P(A) = \dfrac{a}{a+b}$.

Example: In Pokémon FireRed/LeafGreeen, the odds of catching a paralyzed, level 70 Mewtwo with 1 HP if you hurl an Ultra Ball at it is 17 to 583, what is the probability of catching the Mewtwo?

Odds Ratio

The odds ratio is simply a ratio of odds (surprise!). It is usually used to compare two levels of one variable at the same level of another, but is also useful in the interpretation of a logistic regression model.

$$\text{Odds Ratio} = OR = \frac{\text{odds}(A)}{\text{odds}(B)} = \frac{\dfrac{P(A)}{1 - P(A)}}{\dfrac{P(B)}{1 - P(B)}}$$

- If the odds ratio > 1, the odds in the numerator are bigger than the odds in the denominator.

- If the odds ratio < 1, the odds in the denominator are bigger than the odds in the numerator.
- If the odds ratio $= 1$, the two odds are equal.

One thing to be careful of here is that we don't confuse a ratio of odds with a ratio of probabilities. For instance, an odds ratio of 4 does not mean that one event is 4 times more likely to occur than the other. It means that one event has 4 times the odds of the other.

Example: One influential example of using the odds ratio is a study comparing smoking to lung cancer. This was not an easy relationship to discover; it would be highly unethical to perform a controlled experiment studying this relationship. An alternative study type is a "case-control" study. If we want to see if lung cancer is associated with smoking, we can find a bunch of people with lung cancer (the "cases") and a bunch of people without lung cancer (the "controls"). Then we can see how many people in each category smoke, and compute the relevant odds ratio. In 1950, epidemiologists Doll and Hill did such a study and the data are presented below.

		Cancer Status		
		Lung Cancer	No Cancer	Total
Smoking Status	Smoker	1350	1296	2646
	Non-Smoker	7	61	68
	Total	1357	1357	2714

- Note that, in this study, it not meaningful to compute probabilities of people having cancer or not having cancer, because we intentionally sampled the same number of each. We also cannot compute probabilities of being a smoker or not being a smoker, because this is not a genuine random sample: if it is true that smoking causes cancer (spoiler alert: it is), then there will be a greater number of smokers in our sample than there are in the overall population, since we made a point of sampling the same number of "cases" as "controls".
- This is a big advantage that the odds ratio has over simply calculating probabilities. Probabilities calculated from contingency tables are not valid if the row or column totals for the variable of interest are "fixed", i.e. determined by a non-random process.
- Odds ratios, however, compare relative probabilities, and even if the individual probabilities are not meaningful on their own, their relative magnitudes will still be meaningful. So, even though odds ratios are more difficult to interpret, they are sometimes the only way to analyze data collected on categorical variables.

Computing and interpreting the odds ratio for cancer:

Logistic Regression Diagnostics

Just like in simple or multiple linear regression, we should look at diagnostics when doing logistic regression. Most of the diagnostics discussed before are still valid here.

Residuals: Before we discuss the plots, we will first discuss residuals for logistic regression. Since each y value is either 0 or 1, all residuals will be either $1 - \hat{\pi}$ or $-\hat{\pi}$. Those will always simply be parallel lines and not as informative as the residuals in linear regression. There are many different residuals that have been developed for GLMs including: deviance, Pearson, working, response, and partial residuals. Each is useful for different purposes. We will focus on one other type of residual: the standardized Pearson residuals.

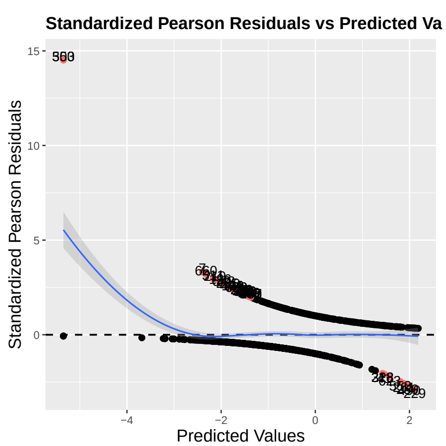
The standardized Pearson residuals (also called the studentized Pearson residuals) are found using

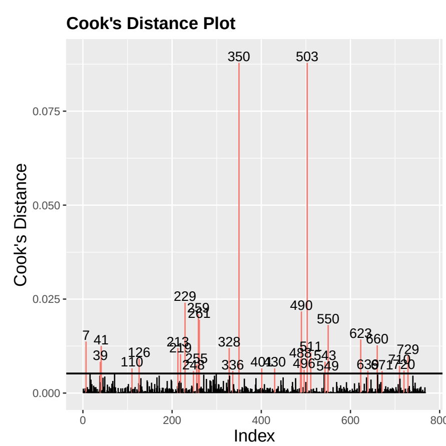
$$r_{SP_i} = \frac{y_i - \hat{\pi}_i}{\sqrt{\hat{\pi}_i(1-\hat{\pi}_i)(1-h_i)}},$$

where h_i is the leverage (or hat) value of the ith observation. It can be shown that these residuals approximately follow a standard normal distribution: $N(0, 1)$.

1. Residual plot: in linear regression, we often plotted the jackknife residuals against the fitted (or predicted) values. We used the `fitted.values()` function to obtain these fitted values. Now we will use the standardized Pearson residuals instead of the jackknife residuals. Instead of the fitted values, it is more common in logistic regression to plot the residuals against the linear predictors, $b_0 + b_1 X_1$, not the predicted probabilities. These are obtained using the `predict()` function in **R**.

   If the model is adequate, then fitted a lowess smoother to the plot should result in an approximately horizontal line with an intercept of zero. On the course web page, the `influence_plots()` function works with logistic regression models as well. The plot is given on the left below.





2. Cook's Distance: Points in logistic regression can still be highly influential. Plotting Cook's distance can be useful in identifying points that influence all of the predicted values. The plot is on the right above.

3. DfFits: Like before, this measures the change in the ith prediction when the ith observation is removed.

4. DfBetas: The jth DfBetas measures the change in b_j when the ith predictor is removed.

We can see in both plots above that observations 350 and 503 are very abnormal and influential. Upon further investigation, we can see those correspond to two individuals who had a mean glucose concentration of 0 and yet tested positive.

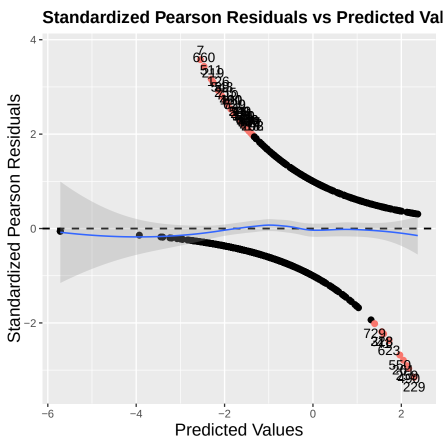
```
pima$glucose[c(350,503)]
```

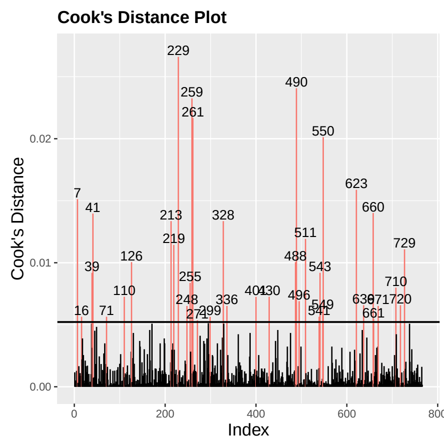
```
[1] 0 0
```

```
pima$test[c(350,503)]
```

```
[1] 1 1
```

Removing these observations and refitting the model yields the following plots with the model





Other Comments on Logistic Regression

1. In **R**, we can obtain the linear predictors, $b_0 + b_1 X_1$, using the `predict` function. Example: `predict(logitmod)`. We can obtain the predicted probabilities, $\hat{\pi}$ using the `fitted.values` function or the `$fitted` command for the model. So `fitted.values(logitmod)` gives the same result as `logitmod$fitted` and `ilogit(predict(logitmod))`. Then we can get our predictions (0 or 1) using `as.numeric(logitmod$fitted > 0.5)`.

2. In general, as many variables as desired may be used as explanatory factors in a logistic regression model. This can be done in the following way:

$$\hat{\pi}(x) = \frac{\exp\left\{b_0 + b_1 x_1 + \cdots + b_p x_p\right\}}{1 + \exp\left\{b_0 + b_1 x_1 + \cdots + b_p x_p\right\}}, \quad \text{for } p \text{ predictors } x_1, \ldots, x_p.$$

   Like linear regression models, logistic regression models can still suffer from multicollinearity (discussed in more detail later) as each slope is interpreted holding the others constant. Namely, $\exp(b_i)$ is the predicted change in the odds of a success when X_i increases by 1, holding all other X_j $j \neq i$ constant.

3. In the coefficients table given by **R**, asymptotic standard errors, a z-test statistic for examining partial significance of the parameters, and the corresponding p-value are given. Without going into the details, the use of the z-statistic is known as the Wald test; it is one of three favored tests for examining parameter significance in this model. In the

example, both parameters are highly significant as their p-values are both $< 2 \times 10^{-16}$. In practice, the recommendation is to not use this Wald test, but rather to use what is known as the **likelihood ratio test**. This can be accomplished by typing

```
anova(logitmod, test = "Chisq")
Analysis of Deviance Table
Model: binomial, link: logit
Response: test
Terms added sequentially (first to last)
        Df  Deviance Resid. Df Resid. Dev   Pr(>Chi)
NULL                       767  993.48391
glucose  1 184.76427       766  808.71964 < 2.22e-16 ***
```

This is testing the hypotheses $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$, the likelihood ratio test statistic is 184.76, which follows a χ^2 distribution (with 1 df in this case). This p-value is also close to zero, indicating that glucose concentration is a highly significant predictor of diabetes incidence. If more variables are in the model, then each will get its own χ^2 test stat and p-value. These are tested sequentially, so order matters.

4. Logistic regression (and multinomial logistic regression) are **classification methods** since they assign observations to groups.

Multinomial Logistic Regression

Logistic regression works well for predicting binary variables – those with only two levels. However, it cannot be used if there are more than two categories. A generalization of logistic regression to work with J different categories is **multinomial logistic regression**. This is also known as polytomous or multicategory logistic regression. This is best used for nominal categorical variables. There are other methods used for ordinal variables.

In regular logistic regression, we found a model for $\log\left[\dfrac{P(Y=1|x)}{1-P(Y=1|x)}\right] = \log\left[\dfrac{P(Y=1|x)}{P(Y=0|x)}\right]$. Calling $Y = 0$ category 1 and $Y = 1$ category 2, we can write $P(Y = 0|x) = \pi_1(x)$ and $P(Y = 1|x) = \pi_2(x)$ and we found $\log\left[\dfrac{\pi_2(x)}{\pi_1(x)}\right] = \beta_0 + \beta_1 x$.

For multinomial logistic regression, we have more categories to consider. For example, suppose we have three groups: $Y = 1$, $Y = 2$ and $Y = 3$. We should obtain three models:

$$\log\left[\frac{\pi_2(x_i)}{\pi_1(x_i)}\right] = \beta_{0,12} + \beta_{1,12}x_i \Longrightarrow \frac{\pi_2(x_i)}{\pi_1(x_i)} = \exp\left(\beta_{0,12} + \beta_{1,12}x_i\right)$$
$$\log\left[\frac{\pi_3(x_i)}{\pi_1(x_i)}\right] = \beta_{0,13} + \beta_{1,13}x_i \Longrightarrow \frac{\pi_3(x_i)}{\pi_1(x_i)} = \exp\left(\beta_{0,13} + \beta_{1,13}x_i\right)$$
$$\log\left[\frac{\pi_3(x_i)}{\pi_2(x_i)}\right] = \beta_{0,23} + \beta_{1,23}x_i \Longrightarrow \frac{\pi_3(x_i)}{\pi_2(x_i)} = \exp\left(\beta_{0,23} + \beta_{1,23}x_i\right)$$

We don't have to actually obtain all three models, because we know that $\pi_1(x_i) + \pi_2(x_i) + \pi_3(x_i) = 1$. So, we can ignore that third model and just use the two that compare π_2 and π_3 to π_1 as the baseline category. Now it's algebra time!

Set up the equations:

$$\frac{\pi_2}{\pi_1} = \exp\left(\beta_{0,12} + \beta_{1,12}x_i\right) \qquad \text{and} \qquad \frac{\pi_3}{\pi_1} = \exp\left(\beta_{0,13} + \beta_{1,13}x_i\right)$$

then

$$\pi_2 = \pi_1 \exp\left(\beta_{0,12} + \beta_{1,12}x_i\right) \qquad \text{and} \qquad \pi_3 = \pi_1 \exp\left(\beta_{0,13} + \beta_{1,13}x_i\right)$$

Using the fact that $\pi_1 + \pi_2 + \pi_3 = 1$ so $\pi_1 = 1 - \pi_2 - \pi_3$:

$$\pi_2 = (1 - \pi_2 - \pi_3)\exp\left(\beta_{0,12} + \beta_{1,12}x_i\right) \quad \text{and} \quad \pi_3 = (1 - \pi_2 - \pi_3)\exp\left(\beta_{0,13} + \beta_{1,13}x_i\right)$$

Now this is a system of two equations and two unknowns. Solving for π_2 and π_3, and then putting hats were appropriate, gives us:

$$\begin{aligned}
\hat{\pi}_2(x_i) &= \frac{\exp(b_{0,12} + b_{1,12}x_i)}{1 + \exp(b_{0,12} + b_{1,12}x_i) + \exp(b_{0,13} + b_{1,13}x_i)}, \\
\hat{\pi}_3(x_i) &= \frac{\exp(b_{0,13} + b_{1,13}x_i)}{1 + \exp(b_{0,12} + b_{1,12}x_i) + \exp(b_{0,13} + b_{1,13}x_i)}, \\
\hat{\pi}_1(x_i) &= 1 - \hat{\pi}_2(x_i) - \hat{\pi}_3(x_i).
\end{aligned}$$

So, in general, if we have J categories,

$$\boxed{\hat{\pi}_j(x_i) = \frac{\exp(b_{0,1j} + b_{1,1j}x_i)}{1 + \sum_{k=2}^{J} \exp(b_{0,1k} + b_{1,1k}x_i)}, \qquad \hat{\pi}_1(x_i) = 1 - \sum_{j=2}^{J} \hat{\pi}_j(x_i)}$$

We will obtain each of the different logistic models using the `multinom()` function from the `nnet` library. By default, it has the "baseline" category (the one we compare the others to) as the first level of a factor variable, the first alphabetically if it is a character, or the smallest number if it is stored as numeric.

Multinomial Logistic Regression Example

A study was undertaken to determine the strength of association between several risk factors and the duration of pregnancies. The risk factors considered were mother's age, nutritional status, history of tobacco use, and history of alcohol use. The response of interest, pregnancy duration, is a three-category variable that was coded as follows: $Y = 1$ indicates preterm (less than 36 weeks), $Y = 2$ indicates intermediate term (36 to 37 weeks), and $Y = 3$ indicates full term (38 weeks or greater). Relevant data for 102 women who had recently given birth at a large metropolitan hospital were obtained. The data can be found in the `pregnancy.txt` file on Canvas.

The response variable, pregnancy duration (y), is in the first column. Nutritional status, is an index of nutritional status (higher score denotes better nutritional status). The predictor variable for age was categorized into different groups: age1 represented those who are less than 20 years of age and age2 are those who were greater than 30 years of age. Everyone else is between 21 and 30 years of age. Alcohol and smoking history were also qualitative predictors; the categories were `Yes" (coded 1) and`No" (coded 0) for each. We will treat this response variable as nominal even though there is a natural ordering. There is another method we can use for ordinal variables that we will not discuss here.

```
library(nnet) # Load the nnet library
preg <- math3150package::pregnancy # Rename the data
preg_mods <- multinom(duration ~ nutrition + age1 + age2 + alcohol + smoking,
                      data = preg) # Fit the multinomial regression model
summary(preg_mods) # Model summary
```

```
Coefficients:
  (Intercept)  nutrition        age1       age2    alcohol    smoking
2   -1.517009 0.01897219 -0.04355881 -0.1721573 -0.9758857 -0.2218629
3   -5.475353 0.06542052 -2.95693397 -2.0596355 -2.0428964 -2.4522828

Std. Errors:
  (Intercept)  nutrition      age1      age2   alcohol   smoking
2    2.047984 0.01631570 0.7373261 0.6943946 0.5869073 0.5856117
3    2.271691 0.01823932 0.9644844 0.8947673 0.7097457 0.7315046

Residual Deviance: 168.6754
AIC: 192.6754
```

We can obtain the p-values for each of our predictors in both models using the fact that $z = b_i/s_{b_i}$ is approximately $N(0,1)$. Using the `multinom_summary()` function in the `math3150package` will give these to us.

```
multinom_summary(preg_mods)
```

```
$`2 vs 1 (baseline)`
 term        estimate std.error  z.stat p.value
 <chr>          <dbl>     <dbl>   <dbl>   <dbl>
1 (Intercept)  -1.52      2.05   -0.741   0.459
2 nutrition     0.0190    0.0163  1.16    0.245
3 age1         -0.0436    0.737  -0.0591  0.953
4 age2         -0.172     0.694  -0.248   0.804
5 alcohol      -0.976     0.587  -1.66    0.0964
6 smoking      -0.222     0.586  -0.379   0.705

$`3 vs 1 (baseline)`
 term        estimate std.error z.stat  p.value
 <chr>          <dbl>     <dbl>  <dbl>    <dbl>
1 (Intercept)  -5.48      2.27    -2.41 0.0159
2 nutrition     0.0654    0.0182   3.59 0.000335
3 age1         -2.96      0.964   -3.07 0.00217
4 age2         -2.06      0.895   -2.30 0.0213
5 alcohol      -2.04      0.710   -2.88 0.00400
6 smoking      -2.45      0.732   -3.35 0.000801
```

Doing a χ^2 test for overall model significance:

```
int_mods <- multinom(duration ~ 1 , data = preg)
anova(int_mods, preg_mods) # Compare full model to intercept model
```

```
Response: duration
      Model Resid. df Resid. Dev   Test    Df LR stat.      Pr(Chi)
1          1       202   220.6862
2 full model       192   168.6754 1 vs 2    10 52.01072  1.13591e-07
```

Of course, there is nothing particularly special about using group 1 as the baseline. We can always change that. If we use group 3 as the reference instead of group 1:

```
duration2 <- factor(preg$duration, levels = c(3, 2, 1))
preg_mods2 <- multinom(duration2 ~ nutrition + age1 + age2 +
                         alcohol + smoking,
                       data = preg, trace = FALSE)
# Putting trace = FALSE silences a message about weights in the model
multinom_summary(preg_mods2)
```

```
$`2 vs 3 (baseline)`
# A tibble: 6 x 5
  term        estimate std.error z.stat  p.value
  <chr>          <dbl>     <dbl>  <dbl>    <dbl>
1 (Intercept)   3.96      1.94     2.04 0.0414
2 nutrition    -0.0464    0.0149  -3.12 0.00181
3 age1          2.91      0.858    3.40 0.000680
4 age2          1.89      0.809    2.33 0.0196
5 alcohol       1.07      0.650    1.64 0.100
6 smoking       2.23      0.668    3.34 0.000844

$`1 vs 3 (baseline)`
# A tibble: 6 x 5
  term        estimate std.error z.stat  p.value
  <chr>          <dbl>     <dbl>  <dbl>    <dbl>
1 (Intercept)   5.48      2.27     2.41 0.0159
2 nutrition    -0.0654    0.0182  -3.59 0.000335
3 age1          2.96      0.964    3.07 0.00217
4 age2          2.06      0.895    2.30 0.0213
5 alcohol       2.04      0.710    2.88 0.00400
6 smoking       2.45      0.732    3.35 0.000801

$AIC
[1] 192.6754

$deviance
[1] 168.6754
```

Interpreting Coefficeints in Multinomial Logistic Regression

We can interpret the model coefficients in terms of **relative risk** instead of odds ratios. In general, relative risk compares the probability of an event occurring in one group compared to the probability of an event occurring in the other group. In this case, we are comparing the predicted probability of belonging to group j for $j = 2, \dots, J$ to belonging to group 1. In particular, as x increases by 1, the relative risk of belonging to group j over group 1 increases by a factor of $\exp(b_{1j})$.

In our example using group 3 as the reference group the slope of nutrition for group 2 is -0.04645 and can be defined as:

Smoking slope: the relative risk of being in the intermediate term pregnancy group compared to the full term group increases by $(\exp(2.2305) - 1) \cdot 100\% = 830\%$ for women with a smoking history.

Remember that does not mean that the women is 8.3 times more likely to have an intermediate term pregnancy compared to before. It just means the ratio in probabilities increases by 8.3 times. For example, it could have been a ratio of 0.3/0.3 for intermediate to full and change to 0.8/0.1. The probability in this case of being in the intermediate group increased by 1.67 times while the probability of being in the full term group fell 2/3rds.

Obtaining Predictions

The predicted group will be the one that has the highest probability. We can use the `fitted.values()` function to obtain all of the predicted probabilities for each group for each observation and we can use the `predict()` function to obtain the predictions. The `predict()` function behaves differently for this `multinomial` object than it did in logistic regression.

```r
predtest <- predict(preg_mods2) # Obtain model predictions
(pred_tab <- table(truth = duration2, prediction = predtest)) # Contingency table
```

```
     prediction
truth  3  2  1
    3 32  5  4
    2  8 23  4
    1  4 10 12
```

```r
sum(diag(pred_tab)) / sum(pred_tab) # Proportion of correct predictions
```

```
[1] 0.6568627
```

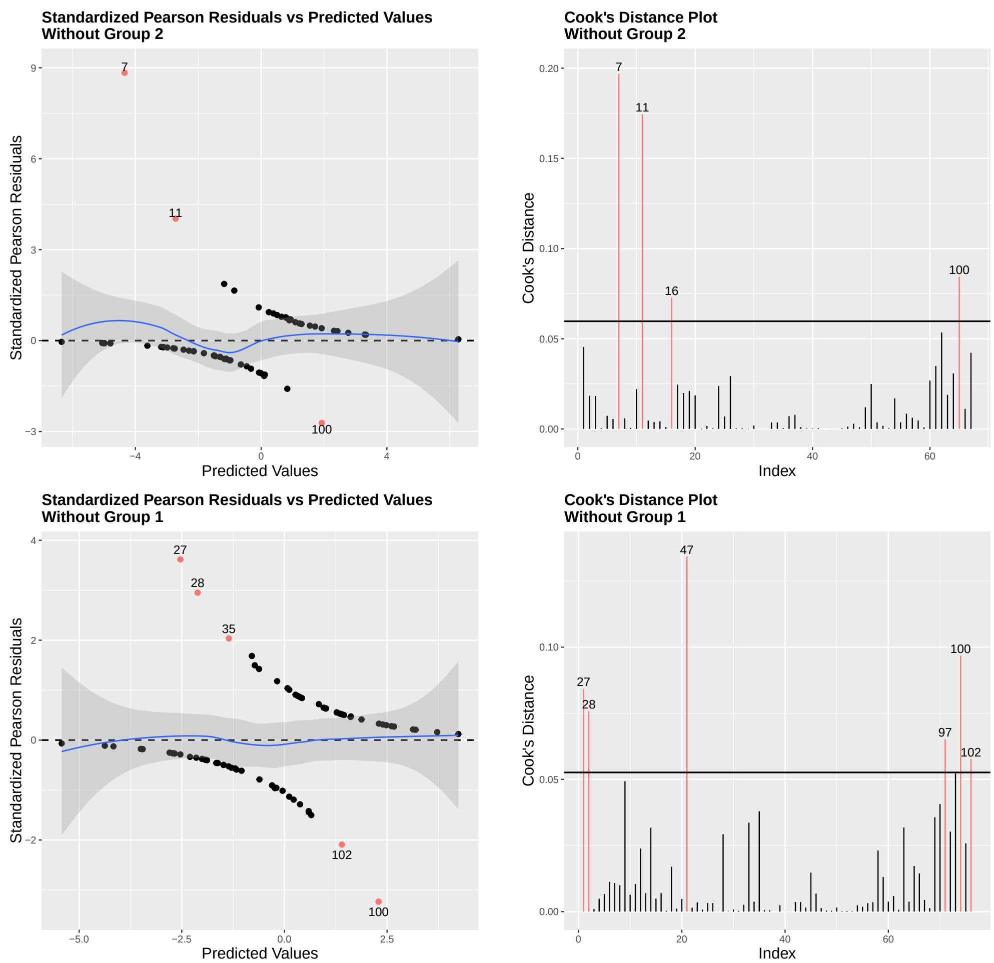
Multinomial Logistic Regression Diagnostics

The best way to check diagnostics for multinomial logistic regression is to look at the $J-1$ individual binary logistic regression plots. In our example above, using group 3 as the baseline, that would be equivalent to fitting two logistic regression models individually: one using the data where only group 2 and group 3 are in the model and the other were only group 1 and group 3 are in the model. Then we can apply the same diagnostics as in binary logistic regression. The function `logistic_plots()` in the `math3150package` can be used for this. This function also relies on the `influence_plots()` function. The input for `logistic_plots()` is a logistic regression model (either a binary or multinomial model).

Example: Returning to the pregnancy example:

```
logistic_plots(preg_mods2)
```

A few of the plots are shown below:



The top graphs are without group 2. The bottom graphs are without group 1. Note that the point number listed is in the original data set. For instance, there are not 100 points in the model when group 2 is removed, but point 100 is flagged since that corresponds to the observation in row 100 in the original data frame.

Bayesian Methods

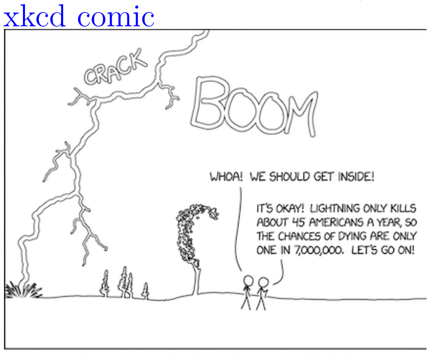
All of the statistical inference you have learned up to this point has been in the **frequentist** framework. In this view, all parameters are fixed numbers and we use methods to try to determine what those numbers are. We do not associate probabilities with parameters, but we can establish the probability that our inferential procedures are correct. All of the hypothesis testing (even the nonparametric tests) and confidence intervals we have dealt with are frequentist inference methods.

However, there is an entire other paradigm that many statisticians have adopted known as the **Bayesian** framework, named after Thomas Bayes who lived in the 18th century and the theorem he developed. In Bayesian statistics, parameters are viewed as random variables that have probability distributions. If we know that the probability distribution is for a parameter, we can simply obtain random samples from that distribution to perform inference. Bayesian statistics is becoming increasingly popular and is especially useful in **uncertainty quantification** in statistics and other fields.

To get an idea of how Bayesian statistics works, let's first discuss conditional probability and Bayes' Theorem.

Conditional Probability

xkcd comic



THE ANNUAL DEATH RATE AMONG PEOPLE WHO KNOW THAT STATISTIC IS ONE IN SIX.

As suggested in the comic to the left, probabilities can change based on the condition that another event has occurred. For example, what is the probability that a randomly chosen man is over 6 feet tall? About 0.14. What is that same probability given that you are selecting from an NBA basketball team? Almost 1. The probability changes given the fact that you are selecting from a basketball team.

Definition: If A & B are any events with $P(B) > 0$, then the **conditional probability of A given that B occurred** is:

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}} \quad \text{"prob of } A \text{ given } B\text{".}$$

Example: Roll fair die 3 times ($6^3 = 216$ total outcomes in S).

Let A = event that sum is 4 or less, so A =

Let B = event that first roll is even. What is $P(A|B)$?

- B can be viewed as a restricted sample space (i.e.: we know the outcome is in B, now what is prob. it is also in A?).

Bayes' Theorem

Useful Result: For any events A and B (where $0 < P(B) \leq 1$):

$$\begin{aligned} \underline{P(B)} &= P((A \cap B) \cup (A^c \cap B)) = P(A \cap B) + P(A^c \cap B) \\ &= \underline{P(B|A)P(A) + P(B|A^c)P(A^c)} \text{ (using the definition of conditional prob.).} \end{aligned}$$

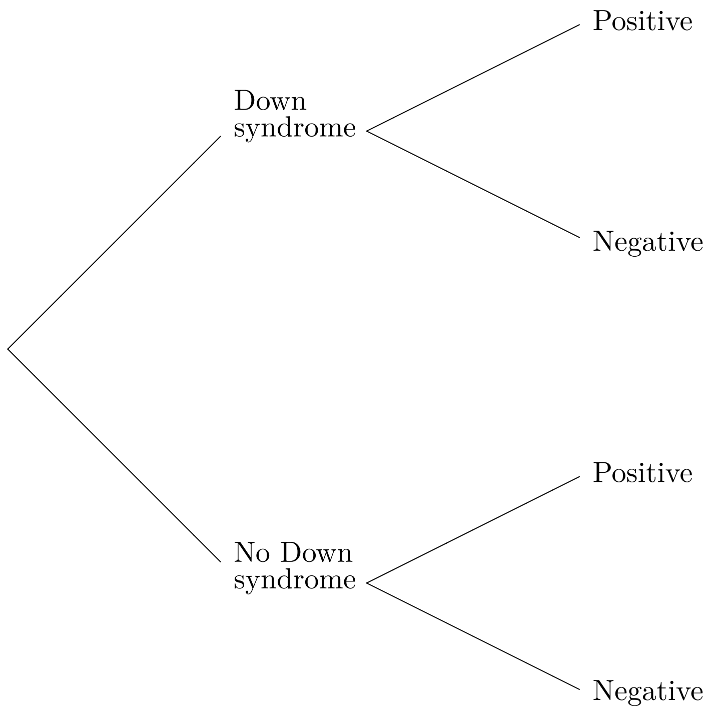
Example: (False positives): For pregnant women over age 35, doctors often recommend a test for Down syndrome early in the second trimester. This is due to a higher risk of having a child with Down syndrome. One test screens for the presence of alpha fetoprotein (AFP) in the mother's blood. This test is somewhat controversial because of its high **false positive** rate.

- When this test is given to a mother of age 36, the probability that:
  - the test is positive when there is a baby with Down syndrome is 0.999,
  - the test is positive when there is not a baby with Down syndrome is 0.05,
  - the test is negative when there is a Down syndrome baby is
  - the test is negative when there is not a Down syndrome baby is

Problem: some women are told they have a Down syndrome baby when they don't (false positive).

- Suppose 1 of every 400 babies born to mothers over the age of 35 have Down syndrome.
- Given that the AFP test is positive, what is the probability a randomly chosen mother of this age has a Down syndrome baby?
- Let:
- Know:
- Want:
- So there is a ______ chance of having a Down syndrome baby when the mother tests positive!

- What happened here? With only 1 in 400 having Down syndrome, in every 400 babies, we expect: 1 to have Down syndrome & nearly 20 to test positive w/o Down syndrome (5%).
- Your chances of being in the false positive group are much higher.
- Another way to solve this problem is with a tree diagram:



General Useful Result: Let B be any event in the sample space S and let $A_1, \ldots, A_n$ form a partition (disjoint) of S, with $P(A_i) > 0$. Then $B = (B \cap A_1) \cup (B \cap A_2) \cup \cdots \cup (B \cap A_n)$, a disjoint union since F_i's are disjoint.

$$\text{So:} \quad \underline{P(B)} = P\left(\bigcup_{i=1}^{n} B \cap A_i\right) = \sum_{i=1}^{n} P(B \cap A_i)$$
$$= \underline{\sum_{i=1}^{n} P(B|A_i)P(A_i)}.$$

Bayes Theorem (in discrete case): Let $A_1, \ldots, A_n$ partition S, so that $P(A_i) > 0$ for $i = 1, \ldots, n$ and let B be any event such that $P(B) > 0$. Then for each $i = 1, \ldots, n$:

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{n} P(B|A_j)P(A_j)}.$$

Example: A hospital receives 40% of its flu vaccine from Company A, 50% from Company B, and the rest from Company C. From Company A, 3% of its vaccine doses are ineffective, from Company B, 2% are ineffective, and from company C, 4% are ineffective. Given a randomly chosen dose is ineffective, what is the probability it came from Company B? Let A, B, C be the events the dose came from Company A,B, or C, respectively and let I be the event the dose is ineffective.

What is $P(A|I) + P(B|I) + P(C|I)$?

Probability Distributions in Bayesian Framework

The Bayes' Theorem we saw earlier was the theorem in the case where we have discrete variables that can be partitioned into different events. Bayes' Theorem also works with continuous probability distributions. In this case, we replace the summation with an integral:

Bayes' Theorem (in continuous case): The distribution for some parameters θ when given data x is

$$p(\theta|x) = \frac{p(\theta)p(x|\theta)}{p(x)} = \frac{p(\theta)p(x|\theta)}{\int p(\theta)p(x|\theta)d\theta} \propto p(\theta)p(x|\theta)$$

- $p(\theta)$ is known as the prior distribution of θ, or just the prior.
- $p(\theta|x)$ is known as the posterior distribution of θ given x, or just the posterior.
- $p(x|\theta)$ is known as the likelihood, which relates all variables in a probability model.
- x and θ are in bold to indicate they are vectors. x is a vector of all of the data values and θ is a vector of all the parameters.
- The symbol "$\propto$" indicates the previous expression is proportional to the next one. That is, it is the same up to a multiplicative constant.

Common Probability Distributions for Bayesian Methods

1. **Normal distribution**: $x|\mu,\sigma \sim N(\mu,\sigma^2)$

$$p(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < \infty$$

   - μ is the mean and σ^2 is the variance.
   - This is a **continuous** probability distribution.

2. **Exponential distribution**: $x|\lambda \sim \text{Exp}(\lambda)$
   - λ is rate parameter. $p(x|\lambda) = \lambda e^{-\lambda x}, \quad x > 0$
   - The mean is $1/\lambda$ and the variance is $1/\lambda^2$.
   - This is a **continuous** probability distribution.

3. **Gamma distribution**: $x|\alpha,\beta \sim \text{Gam}(\alpha,\beta)$

$$p(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

   - α is the shape parameter and β is rate parameter.
   - The mean is α/β and the variance is α/β^2.
   - This is a **continuous** probability distribution.

4. **Uniform distribution**: $x|a,b \sim \text{Unif}(a,b)$

$$p(x|a,b) = \frac{1}{b-a}, \quad a < x < b$$

   - a is the lower bound on the interval and b is the upper bound.
   - The mean is $(a+b)/2$ and the variance is $(b-a)^2/12$.
   - This is a **continuous** probability distribution.

5. **Binomial distribution**: $x|n,p \sim \text{Binom}(n,p)$

$$p(x|n,p) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n$$

   - n is the number of trials and p is the probability of success on each trial.
   - The mean is np and the variance is $np(1-p)$.
   - This is a **discrete** probability distribution.

6. **Poisson distribution**: $x|\lambda \sim \text{Pois}(\lambda)$

$$p(x|n,p) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 1, 2, 3 \ldots$$

   - λ is the average number of events in a time interval, area, or volume.
   - The mean is λ and the variance is λ.
   - This is a **discrete** probability distribution.

Example: Suppose we have 5 data values $x = (1, 5, 3, 2, 4)$ that each come from a normal distribution with a variance of 2.5 and an unknown mean, μ: $x_i \sim N(\mu, 2.5)$.

- In the frequentist framework, we can construct a confidence interval for μ using $\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n} = 3 \pm 1.96 \cdot \sqrt{2.5/5} \implies 1.614 < \mu < 4.386.$
- In the Bayesian framework, we want to obtain the **posterior distribution** of μ, which we write as $p(\mu|x, \sigma^2)$, and then sample from that. To do that, we will first need a **prior distribution** on μ. Let's just assign a uniform prior: $\mu \sim \text{Unif}(-1000, 1000)$ so $p(\mu) = 1/2000$. This is equivalent to saying we have no information about μ (it could be anywhere between -1000 and 1000).

  Now, let's obtain the **likelihood function**: $p(x|\mu, \sigma^2)$. Since we know $x_i \sim N(\mu, 2.5)$, we have $p(x_i|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \mu)^2\right\}$. Then

$$\begin{aligned}
p(x|\mu, \sigma^2) &\propto p(x_1|\mu, \sigma) \times \cdots \times p(x_5|\mu, \sigma) \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_1 - \mu)^2\right\} \times \cdots \times \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_5 - \mu)^2\right\} \\
&= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^5 \exp\left\{-\frac{1}{2\sigma^2}(x_1 - \mu)^2 - \frac{1}{2\sigma^2}(x_2 - \mu)^2 - \cdots - \frac{1}{2\sigma^2}(x_5 - \mu)^2\right\} \\
&= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^5 \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{5}(x_i - \mu)^2\right\}.
\end{aligned}$$

  Now we can obtain the posterior by multiplying the prior and likelihood functions:

$$\begin{aligned}
p(\mu|x, \sigma^2) &\propto p(\mu)p(x|\mu, \sigma^2) \\
&\propto \frac{1}{2000} \cdot \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^5 \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{5}(x_i - \mu)^2\right\} \\
&\propto
\end{aligned}$$

- Once we have the posterior distribution for μ, which is $N(\bar{x}, \sigma^2/n)$, we can sample from that distribution. These samples are all different values of μ.

```
set.seed(2026)
x <- c(1,5,3,2,4)
mu <- rnorm(n = 500000, mean = mean(x), sd = sqrt(2.5/5))
quantile(mu, c(0.025, 0.975))
    2.5%    97.5%
1.612910 4.385999
```

- Taking the 2.5th and 97.5th percentiles gives us a 95% **credible interval** for the population mean, μ. A credible interval is an interval within which an unobserved parameter value falls with a particular probability.
  - In this case, there is a 95% chance that the population mean, μ, lies between 1.613 and 4.386.
  - Compare this to the confidence interval on the previous page.
- What if we used a different prior distribution instead. For example, if we had reason to believe that the true mean was around 6 (and we got a bit unlucky with our sample). We could assign a prior $\mu \sim N(6, \sigma_0^2)$. We will change the value of σ_0^2 to see what affect that has. σ_0^2 is called a **hyperparameter** since it is a parameter in the prior. In this case,

$$p(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left\{-\frac{1}{2\sigma_0^2}(\mu-6)^2\right\}$$

Our posterior is

$$\begin{aligned}
p(\mu|x,\sigma^2) &\propto p(\mu)p(x|\mu,\sigma^2) \\
&\propto \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left\{-\frac{1}{2\sigma_0^2}(\mu-6)^2\right\} \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^5 \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{5}(x_i-\mu)^2\right\} \\
&\propto \exp\left\{-\frac{1}{2\sigma_0^2}(\mu-6)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{5}(x_i-\mu)^2\right\} \\
&\propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}\right)\left(\mu - \frac{1}{\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}}\left[\frac{6}{\sigma_0^2}+\frac{n\bar{x}}{\sigma^2}\right]\right)^2\right\}
\end{aligned}$$

So, the posterior is again normal. In particular, $\mu|x,\sigma^2 \sim N\left(\frac{1}{\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}}\left[\frac{6}{\sigma_0^2}+\frac{n\bar{x}}{\sigma^2}\right], \frac{1}{\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}}\right)$. See the results when $\sigma_0^2 = 1$ and when $\sigma_0^2 = 10$:

```
sigma_sq0 <- 1
mean_mu <- 1/(1/sigma_sq0 + 5/2.5) * (6/sigma_sq0 + 5*mean(x)/2.5)
sigma_mu <- sqrt(1/(1/sigma_sq0 + 5/2.5))
mu <- rnorm(500000, mean_mu, sigma_mu)
quantile(mu, c(0.025, 0.975))
    2.5%    97.5%
2.869030 5.129936
```

```r
sigma_sq0 <- 10
mean_mu <- 1/(1/sigma_sq0 + 5/2.5) * (6/sigma_sq0 + 5*mean(x)/2.5)
sigma_mu <- sqrt(1/(1/sigma_sq0 + 5/2.5))
mu <- rnorm(500000, mean_mu, sigma_mu)
quantile(mu, c(0.025, 0.975))
```

```
    2.5%    97.5%
1.792919 4.492821
```

Summary:

- We assign a prior distribution to the parameter we are trying to estimate.
- We assign a likelihood function to the data based on what we assume the population to look like or what distribution fits our data best.
- We obtain a function that is proportional to the posterior for our parameter by multiplying the prior and likelihood together.
- We identify what distribution has a kernel like the one we see in our posterior (thinking of our parameter now as x_i).
- We obtain lots of samples from that posterior distribution.
- We take the $\frac{\alpha}{2}th$ and $(1-\frac{\alpha}{2})th$ quantiles to obtain the $(1-\alpha)100\%$ credible intervals.
  - That credible interval is the Bayesian version of a confidence interval and is used for inference on the parameter.

MCMC Sampling

MCMC stands for Markov chain Monte Carlo and MCMC methods are a class of algorithms for sampling from a probability distribution. Monte Carlo sampling is the process of relying on many random samples to obtain numerical results and a **Markov chain** is a process in which the next event depends only on the previous event. There are two primary MCMC sampling methods: Gibbs sampling and Metropolis sampling.

Gibbs Sampling

The basic Gibbs sampler is used in much the same way as we saw on the previous few pages. We sample directly from a posterior distribution to obtain parameter values. With Gibbs sampling, however, we can sample from more than one distribution at a time while updating our parameter values as we go. The steps are shown below with two parameters, θ_1 and θ_2, but this can scale to any number of parameters.

1. Write down the posterior distributions for both θ_1 and θ_2: $p(\theta_1|x, \theta_2)$ and $p(\theta_2|x, \theta_1)$.
2. Select starting values for θ_1 and θ_2 called $\theta_1^{(0)}$ and $\theta_2^{(0)}$.
3. Generate, in turn,
$$\theta_1^{(i+1)} \text{ from } p\left(\theta_1|x, \theta_2^{(i)}\right)$$
$$\theta_2^{(i+1)} \text{ from } p\left(\theta_2|x, \theta_1^{(i+1)}\right)$$
4. Increment i and return to step 3. Continue until you obtain the number of desired samples.

Example: Assume a normal likelihood function with: $x_i|\mu, \sigma^2 \sim N(\mu, \sigma^2)$. Let μ have a normal prior with $\mu \sim N(25, 100)$ and let the precision $\tau = 1/\sigma^2$ have a gamma prior with $\tau \sim \text{Gam}(5, 1/2)$. Additionally, let `x = (23.3, 14.5, 19.0, 20.2, 5.4, 23.8, 21.3, 12.8, 17.6, 19.8, 11.0, 21.5, 15.7, 22.1, 21.0, 13.7, 14.9, 13.8, 17.1, 11.3)`.

In this case, $n = 20$, $\sigma_0^2 = 100$, $\mu_0 = 25$, $\alpha_0 = 5$, $\beta_0 = 1/2$, and $\overline{x} = 16.99$.

Step 1: If we were to go through all the math here, we would get:

$$\mu|x, \tau \sim N\left(\frac{1}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}}\left[\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2}\right], \frac{1}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}}\right) = N\left(\frac{1}{\frac{1}{100} + 20\tau}\left[\frac{25}{100} + 339.8\tau\right], \frac{1}{\frac{1}{100} + 20\tau}\right)$$

and

$$\tau|x, \mu \sim \text{Gam}\left(\alpha_0 + n/2, \beta_0 + \frac{\sum_{i=1}^n (x_i - \mu)^2}{2}\right) = \text{Gam}\left(15, \frac{1}{2} + \frac{\sum_{i=1}^n (x_i - \mu)^2}{2}\right)$$

Step 2: The starting values usually do not matter too much. I will start each at the mean of their prior: $\mu^{(0)} = 25$ and $\tau^{(0)} = 10$.

Step 3: Generate each of the following:

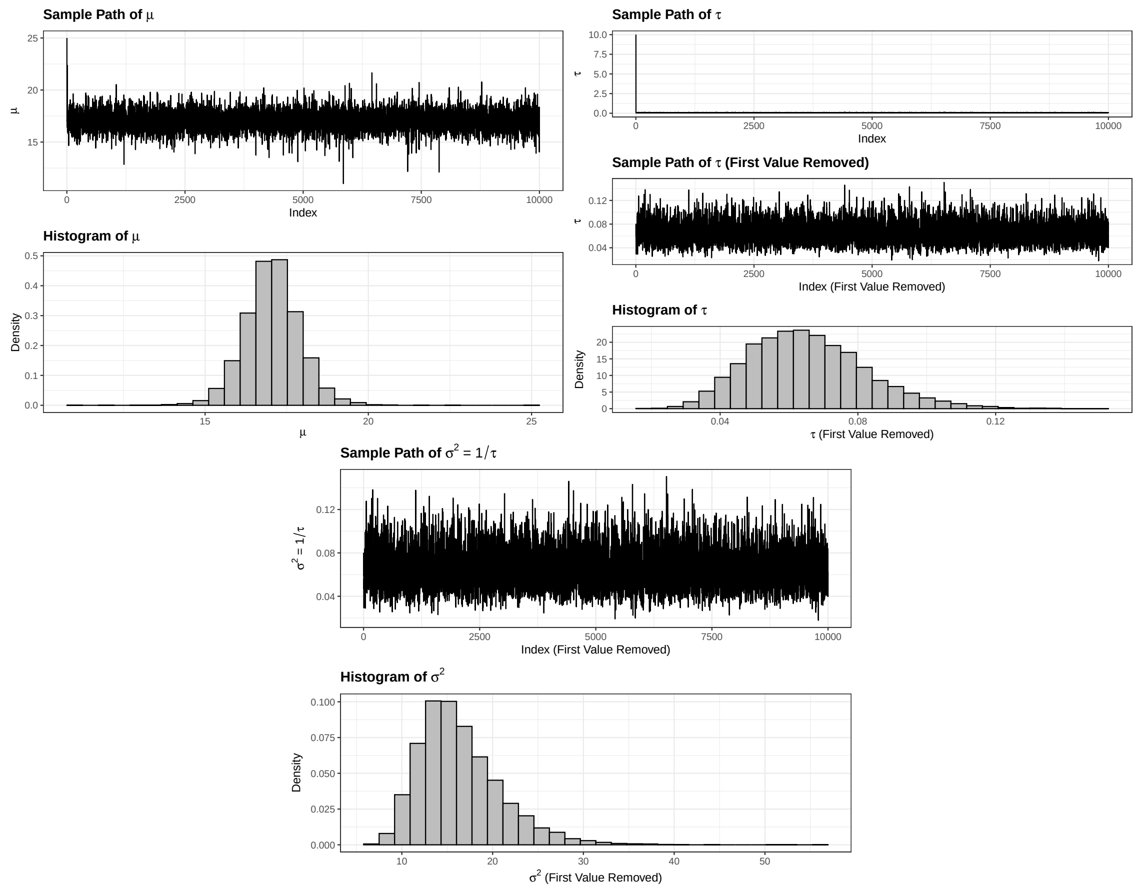
$$\mu^{(i+1)} \text{ from } N\left(\frac{1}{\frac{1}{100}+20\tau^{(i)}}\left[\frac{25}{100}+339.8\tau^{(i)}\right], \frac{1}{\frac{1}{100}+20\tau^{(i)}}\right)$$

$$\tau^{(i+1)} \text{ from Gam}\left(15, \frac{1}{2}+\frac{\sum_{i=1}^{n}(x_i-\mu^{(i+1)})^2}{2}\right)$$

Step 4: Repeat for 10,000 samples.

The **R** code is shown below:

```
x <- c(23.3, 14.5, 19.0, 20.2,  5.4, 23.8, 21.3, 12.8, 17.6, 19.8,
       11.0, 21.5, 15.7, 22.1, 21.0, 13.7, 14.9, 13.8, 17.1, 11.3)
nsamps <- 10000
mu <- rep(0, nsamps)  # Initialize the mu vector
mu[1] <- 25           # Starting value of mu
tau <- rep(0, nsamps) # Initialize the tau vector
tau[1] <- 10          # Starting value of tau
for(i in 1:(nsamps - 1)) {
  mu[i+1] <- rnorm(n = 1, mean = 1/(1/100 + 20*tau[i]) * (25/100 + 339.8*tau[i]),
                   sd = 1/(1/100 + 20*tau[i]))
  tau[i+1] = rgamma(n = 1, shape = 15, rate = 1/2 + 1/2 * sum((x - mu[i+1])^2))
}
```



If the posterior distributions for our parameters are known, but they depend on the other parameter(s), then Gibbs sampling is appropriate. What if we don't know the exact posterior distribution? If we only know it up to a proportionality constant, then **Metropolis** sampling should be used. I will omit the details here, but this the most common type of sampling method.

The downside is that Metropolis sampling is much more complicated and can lead to sampling chains that don't mix well without some tuning.

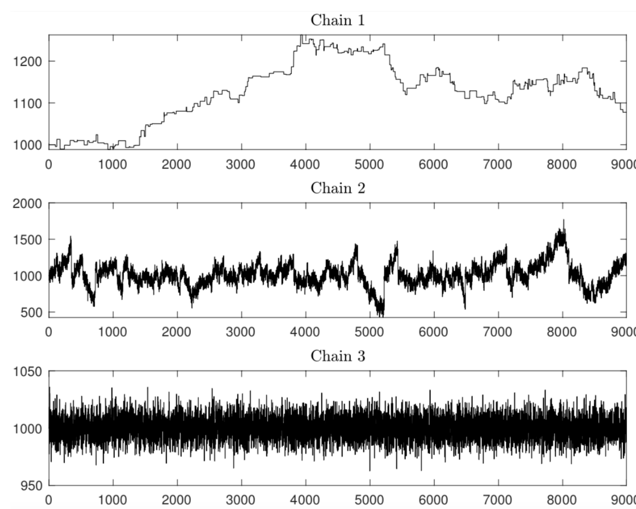
MCMC Diagnostics

There are two important diagnostics we will look at when doing MCMC sampling. Both of which have to do with how well the sampling chains mix.

1. Plots of the sampling chain.
2. The effective sample size.
3. The R-hat statistic.

Plots of the sampling chain(s)

The primary diagnostic tool for evaluating MCMC sampling is observing the sampling chains. Ideally, each sample in the chain will be independent from all others. Since the posterior distributions rely on the previous iteration, that is not going to be the case, but we would like to minimize the number of previous iterations on which the current one depends. A good-looking sampling chain is one that essentially looks like noise, with no discernible patterns or paths.



Effective Sample Size

An important diagnostic that takes into account how correlated the chain is with itself is the effective sample size (ESS). The closer the ESS is to the number of samples in the chain, the better. However, it will usually be less. We want to make sure the ESS is large enough to support the kind of credible intervals we will be creating. Typically, an ESS of 1000 or more is good when making 95% credible intervals.

R-hat

The R-hat statistic is a measure of whether different sampling chains for the same parameter are converging to the same distribution. If they are not converging, then that indicates that the true distribution is likely not being sampled correctly. We can find R-hat by taking $\widehat{R} = \sqrt{\frac{\text{Variance Between Chains}}{\text{Variance Within Chains}}}$. If the variance between chains is larger than the variance within chains, we have issues. We like R-hat to be as close to 1 as possible (less than 1.05). Larger than 1.05 indicates there are likely some issues with convergence.

Bayesian Methods Applied to Regression in R

In **R**, there is a package called `brms` that implements Bayesian methods for regression models using the `brm()` function. BRM stands for Bayesian Regression Models. It works much like the `lm()` and `glm()` functions.

Recall the model predicting the divorce rate in Maine from amount of margarine consumed:

```
maine_mod <- lm(divorce ~ margarine, data = maine_data)
summary(maine_mod)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.308626   0.048032   68.88 2.20e-12 ***
margarine   0.201386   0.008735   23.05 1.33e-08 ***

Residual standard error: 0.03841 on 8 degrees of freedom
```

Now let's use the `brm()` function. By default, it will set priors itself and take 2000 samples across four different sampling chains. It will discard the first 1000 samples from each chain as "warm-up" samples. This will leave us with 4,000 samples.

```
library(brms)
maine_bayes_1 <- brm(divorce ~ margarine, data = maine_data, family = gaussian())
```

There are other, faster back ends we can use. Like `cmdstanr`. This can be installed by running the following code.

```
install.packages(
  "cmdstanr",
  repos = c('https://stan-dev.r-universe.dev',
            getOption("repos"))
  )
cmdstanr::install_cmdstan()
```

This `cmdstanr` back end is generally better, so let's use it from now on.

```
set.seed(1)
maine_bayes <- brm(divorce ~ margarine, data = maine_data,
                   family = gaussian(), backend = "cmdstanr")
summary(maine_bayes)
```

```
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     3.31      0.06     3.19     3.44 1.00     2364     1842
margarine     0.20      0.01     0.18     0.22 1.00     2475     1785

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.05      0.02     0.03     0.09 1.00     1830     1800
```

That `Rhat` value indicates whether the chain is mixing well. A value close to 1 (less than 1.05) indicates that the chains have converged (a good thing!). `Bulk_ESS` is the effective sample size for the estimate while `Tail_ESS` is the effective sample size for the lower and upper credible interval bounds. We want those each to be at least 1000, but the higher the better.

Setting Priors

However, we can also set our own priors on the coefficients. It is the slope and intercept that we are setting the priors for here. The true intercept is called β_0 and the true slope is called β_1. So, if we had reason to believe the true intercept should be 5 (or closer to 5 than our data suggest), we could put a prior centered at 5 with a small variance. Like $\beta_0 \sim N(5, 1)$. Then let's use this prior for β_1: $\beta_1 \sim N(0, 0.1^2)$. We can also put a prior on σ_ϵ. Let's just put a uniform prior on that. In **R**, we can set these priors like this:

```r
maine_priors <- c(
  prior(normal(5, 1), class = "b", coef = "Intercept"),
  prior(normal(0, 0.1), class = "b", coef = "margarine"),
  prior("", class = "sigma")
)
set.seed(1)
maine_bayes_priors <- brm(divorce ~ 0 + Intercept + margarine,
                          data = maine_data, family = gaussian(),
                          backend = "cmdstanr", prior = maine_priors)
summary(maine_bayes_priors)
```

```
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     3.33      0.06     3.22     3.46 1.01      853     1073
margarine     0.20      0.01     0.17     0.22 1.01      850     1107

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.05      0.01     0.03     0.08 1.00     1191     1612
```

Notice that if we put a prior on the intercept, we need to use slightly different syntax: `0 + Intercept`. If using the default prior on the intercept, that is not needed.

We see now that the intercept is closer to 5 (3.33 vs 3.31) and the slope is closer to 0 (0.198 vs 0.201, although they both round to 0.20) while the sigma term dropped slightly as well. If we put stricter priors, they would get even closer to the mean of their priors:

```r
maine_priors_tight <- c(
  prior(normal(5, 0.1), class = "b", coef = "Intercept"),
  prior(normal(0, 0.05), class = "b", coef = "margarine"),
  prior("", class = "sigma")
)
set.seed(1)
```

```
maine_bayes_priors_tight <- brm(
  divorce ~ 0 + Intercept + margarine, data = maine_data,
  family = gaussian(), backend = "cmdstanr", prior = maine_priors_tight
  )
summary(maine_bayes_priors_tight)
```

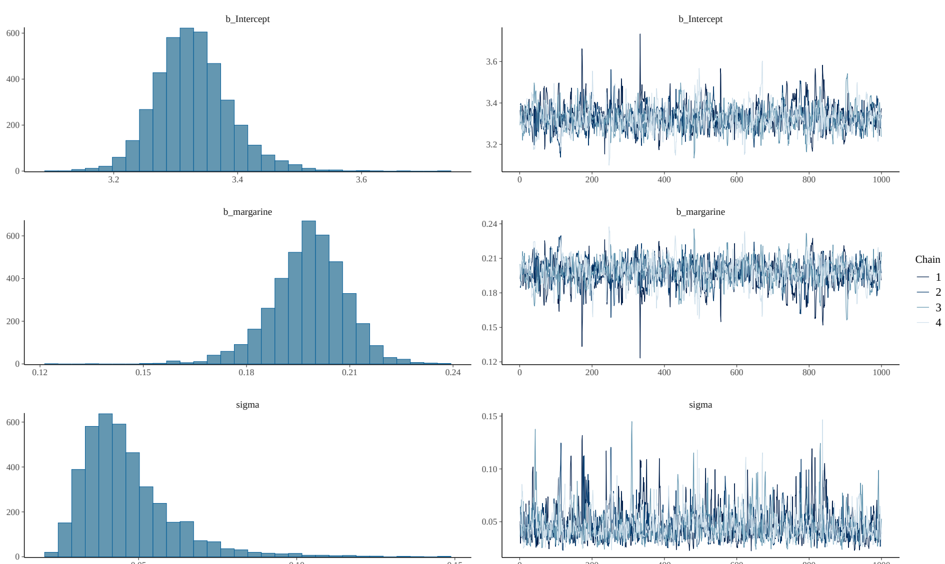
```
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     5.01      0.10     4.82     5.22 1.00     3731     3017
diameter      0.04      0.05    -0.06     0.14 1.00     3434     2877

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    20.42      3.07    15.43    27.52 1.00     3446     2787
```

Notice that sigma got much bigger. The sigma term is the estimate for σ_ϵ, the standard deviation of the residuals. This model has much more error since we are forcing an intercept and slope that don't really align with the data. Therefore, the residuals are getting larger and have a bigger error.

Plots

Let's return to the model with the first priors we set. We can plot the sampling chains using the `plot()` function:

```
plot(maine_bayes_priors)
```



Those chains look pretty good, but could be better. The Rhat is 1.01, and the ESS are close to, or a little under, 1000! That points to the Bayesian modeling performing pretty well, but it would be better to increase the number of samples. We could do this with the `iter` option as is shown on the next page.

```
maine_bayes_priors <- brm(divorce ~ 0 + Intercept + margarine,
                          data = maine_data, family = gaussian(),
                          backend = "cmdstanr", prior = maine_priors,
                          iter = 4000)
summary(maine_bayes_priors)
```

```
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     3.33      0.07     3.21     3.47 1.00     1703     1609
margarine     0.20      0.01     0.17     0.22 1.00     1733     1763

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.05      0.02     0.03     0.09 1.00     1999     1647
```

This looks better!

Math Behind Getting the Posterior

Our goal is to find the posterior of our model parameters given our data. That is $P(\beta_0, \beta_1, \sigma_\varepsilon | \boldsymbol{y}, \boldsymbol{x})$. By putting `family = gaussian()`, we assume a normal distribution for y with a mean of $\beta_0 + \beta_1 x_i$ and a variance of σ_ε^2. That means the likelihood for one observation of Y, y_i can be written $y_i | \beta_0, \beta_1, \sigma_\varepsilon, x_i \sim N(\beta_0 + \beta_1 x_i, \sigma_\varepsilon^2)$. Or, we can write

$$P(y_i | \beta_0, \beta_1, \sigma_\varepsilon, x_i) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left(-\frac{1}{2\sigma_\varepsilon^2}(y_i - \beta_0 - \beta_1 x_i)^2\right)$$

But we need the likelihood for all the y values. We can get this by multiplying them all together. This assumes the values are independent.

$$\begin{aligned} P(\boldsymbol{y} | \beta_0, \beta_1, \sigma_\varepsilon, \boldsymbol{x}) &= \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left(-\frac{1}{2\sigma_\varepsilon^2}(y_1 - \beta_0 - \beta_1 x_1)^2\right) \times \cdots \times \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left(-\frac{1}{2\sigma_\varepsilon^2}(y_n - \beta_0 - \beta_1 x_n)^2\right) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}}\right)^n \exp\left(-\frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2\right). \end{aligned}$$

Then we have normal priors on β_0 and β_1. $\beta_0 \sim N(0, 0.5^2)$ and $\beta_1 \sim N(1, 5^2)$:

$$P(\beta_0) = \frac{1}{\sqrt{2\pi \cdot 0.5^2}} \exp\left(-\frac{1}{2 \cdot 0.5^2}\beta_0^2\right)$$

and

$$P(\beta_1) = \frac{1}{\sqrt{2\pi \cdot 5^2}} \exp\left(-\frac{1}{2 \cdot 5^2}(\beta_1 - 1)^2\right).$$

The prior on sigma is flat, so we can ignore it and $\boldsymbol{x}$ does not have a prior since we think of it as being fixed. That gives the posterior as

$$\begin{aligned}
P(\beta_0, \beta_1, \sigma_\varepsilon | \boldsymbol{y}, \boldsymbol{x}) &\propto P(\boldsymbol{y}|\beta_0, \beta_1, \sigma_\varepsilon, \boldsymbol{x}) \times P(\beta_0) \times P(\beta_1) \\
&= \left(\frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}}\right)^n \exp\left(-\frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right) \\
&\qquad \times \frac{1}{\sqrt{2\pi \cdot 0.5^2}} \exp\left(-\frac{1}{2 \cdot 0.5^2}\beta_0^2\right) \times \frac{1}{\sqrt{2\pi \cdot 5^2}} \exp\left(-\frac{1}{2 \cdot 5^2}(\beta_1 - 1)^2\right) \\
&\propto \frac{1}{\sigma_\varepsilon^n} \exp\left(-\frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 - \frac{1}{2 \cdot 0.5^2}\beta_0^2 - \frac{1}{2 \cdot 5^2}(\beta_1 - 1)^2\right).
\end{aligned}$$

Where the last line is obtained by dropping all the terms we are multiplying (or dividing) by that do not contain β_0, β_1, or σ_ε.

This is the posterior that the `brm()` function is sampling from to get the chains for the intercept, slope of diameter, and sigma that are shown on the previous page. The function automatically tunes things so the sample chains (hopefully) look good.

Bayesian Logistic Regression

Now let's apply the `brm()` function to a logistic regression problem. Recall the Pima diabetes example:

```r
library(faraway)
logitmod <- glm(test ~ glucose, data = pima, family = binomial())
summary(logitmod) # Summary of logistic fit
```

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.350080   0.420827  -12.71   <2e-16 ***
glucose      0.037873   0.003252   11.65   <2e-16 ***
```

When using `brm()`, we need to specify `family = bernoulli()` rather than `binomial()`.

```r
set.seed(1)
logit_bayes <- brm(test ~ glucose, data = pima,
                   family = bernoulli(), backend = "cmdstanr")
summary(logit_bayes) # Summary of logistic fit
```

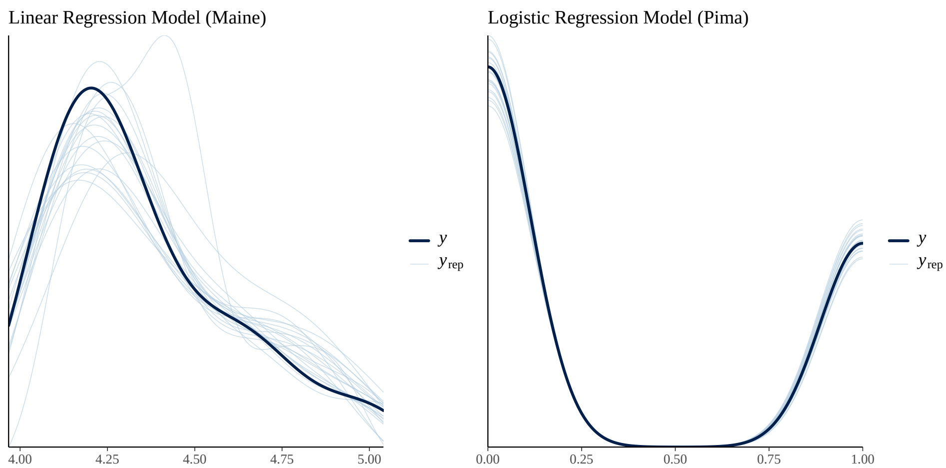
```
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -5.38      0.42    -6.21    -4.58 1.00     3213     2770
glucose       0.04      0.00     0.03     0.04 1.00     3735     2862
```

This is using default priors. We can set priors for this in the same way we did for the other Bayesian models.

Bayesian Diagnostics

In addition to the diagnostics we discussed earlier in this set of notes (residual plots, QQ plots, influence plots, sampling chain plots, effective sample size, R-hat, etc.), it is best to check if we can replicate our data using our model. One very helpful function for this is `pp_check()`.

```
library(patchwork) # Needed to add the plots together on the same graph
set.seed(1)
p1 <- pp_check(maine_bayes_priors, ndraws = 20) +
  labs(title = "Linear Regression Model (Maine)")
p2 <- pp_check(logit_bayes, ndraws = 20) +
  labs(title = "Logistic Regression Model (Pima)")
p1 | p2
```



Ideally, the y_{rep} curves would look similar to the y curve on these plots. We can see that the logistic regression model is looking great. The linear model is looking okay, but not as good.